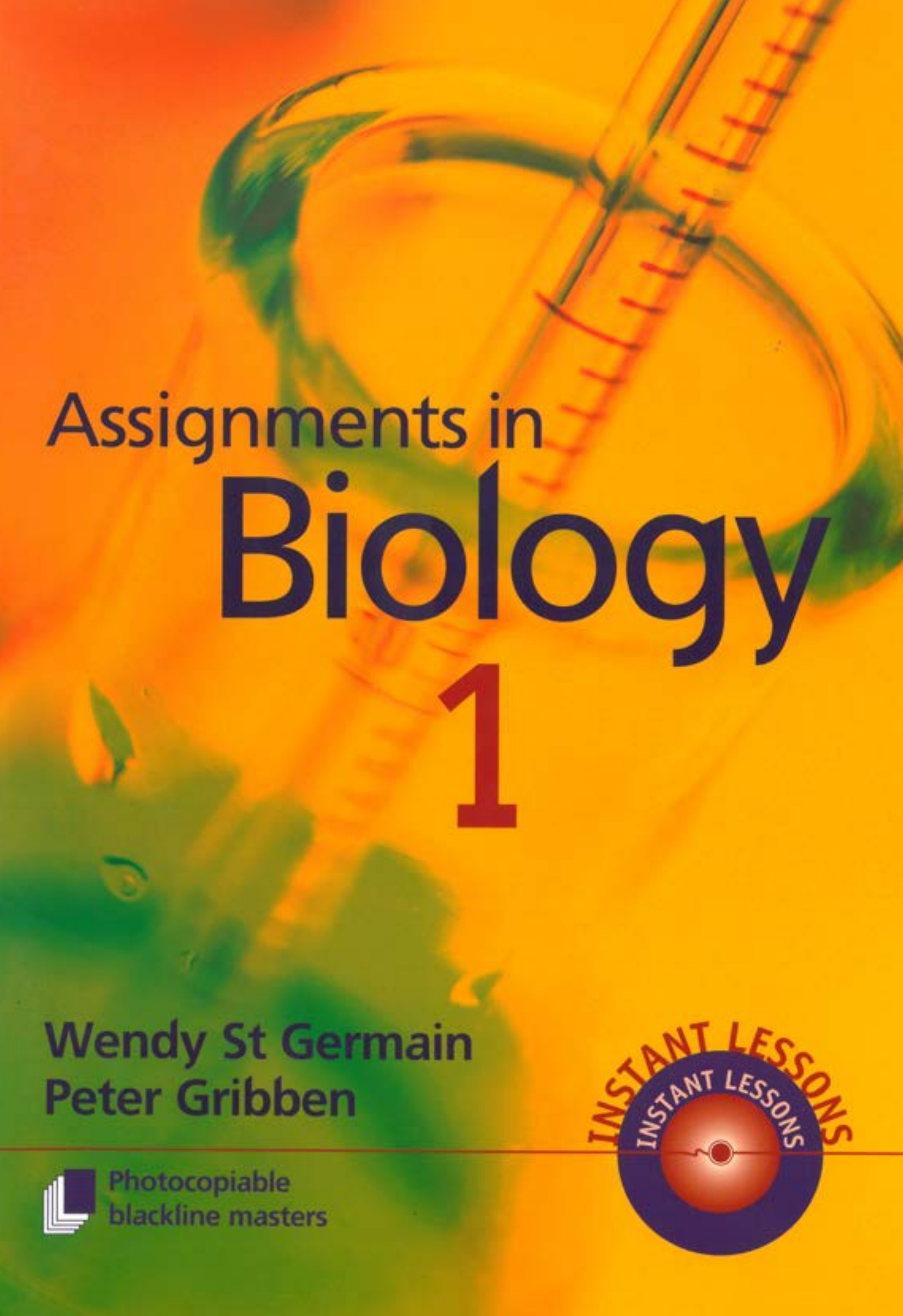




Assignments in Biology 1

Wendy St Germain
Peter Gribben



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Wendy St Germain
Peter Gribben



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Preface

This set of books has been designed to help teachers of biology in most aspects of this subject. Students will find the language, concepts and material much easier to use and assimilate than that in conventional textbooks. The content includes 'stand alone' material as well as sheets that require some existing knowledge and that test these concepts. An understanding of key phrases and concepts is encouraged.

A variety of styles has been used to engage students' attention. Whilst the books have been mainly aimed at senior students, teachers of able year nine and ten students will find much that they can use with these year groups. In order to extend the more motivated students, many topics include a research section. Outline answers are given for these research questions.

The material can be used as reinforcement, revision, homework, and as an introduction to new topics. Replacement teachers who may not be totally familiar with the subject will find these books invaluable.

Note: In most cases space has not been provided for students to answer the questions on the worksheets. Answers should be written in students' usual class work book or folder.

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Classification I: Plants and animals

1

With so many organisms in our world it is important that we group them according to specific features. Classification, or *taxonomy*, is the grouping together of living things according to characteristics that they share which describe and identify them. How do we know if a thing is living or not? Living things have these features in common: they all move, grow, reproduce, breathe, gain energy from food and give off waste products.

Living things also have cells. Cells are the building blocks of life. Some very simple living things are unicellular (have only a single cell), most are multicellular (have many cells all working together). There are many ways to classify things and because of this, an international standard was developed.

Classification begins at the largest grouping, the Kingdom. All species fall under one of two major Kingdoms; plants or animals. The difference between plants and animals is based on three key areas:

- **Movement:** Most animals can move easily. They have special parts to help them move, for example wings, fins, tails, legs etc. Exceptions occur in sessile (non-moving) species, such as the 'flower animals', eg sea anemones (class Anthozoa) and some sea cucumbers (class Holothuroidea). The movement of plants is much less and usually very slow compared to animals. Flowers can open and close and leaves can follow the sun as it appears to move across the sky. Roots of plants grow downwards due to the effect of gravity. These responses by plants to stimuli are called *tropisms*.
- **How they obtain food:** Animals are *heterotrophs*, they can't make their own food. They have to eat plants (herbivores) or other animals (carnivores) or both (omnivores). Plants are *autotrophs*, they make their own food (initially glucose) by photosynthesis. They use their leaves to take in energy from the sun, carbon dioxide from the air and take in water through their roots.
- **Differences in the structure of their cells:** Animal cells have a cell membrane, nucleus, cytoplasm, mitochondria, endoplasmic reticulum and vacuoles. In addition to those features, plants have a rigid cell wall and green plants have chloroplasts.

- 1 Why do we need to classify organisms?
- 2 Give a definition for *phototropism*.
- 3 Give a definition for *geotropism*.
- 4 Name five characteristics of living things.
- 5 Write a word equation for the process of photosynthesis.
- 6 Draw up a table showing the organelles (structures) present in a typical animal cell and a typical green plant cell.
- 7 Draw up a table with the headings: autotroph, heterotroph, herbivore, omnivore and carnivore. Place the following organisms under the appropriate headings. (Note some may appear in more than one category).
grass, cow, sheep, human, fox, tree, rabbit, wattle, kangaroo
Add one more example to each category.

The most specific of the classification groupings is the *species* level, but there are several divisions before reaching this level. From broadest to most specific the divisions are: Kingdom, Phylum, Class, Order, Genus, Species. The closer two organisms are to the species level, the more closely related they are in evolutionary time.

Hint: The mnemonic *Kindly Prevent Confusion Over Grouping Species* will help you to remember the order of the groupings, KPCOGS or Kingdom, Phylum etc.

- 8 Make up your own mnemonic for Kingdom, Phylum, Class, Order, Genus, Species.

One way that is used to help classify animals is to divide them into those with an internal backbone (vertebrates) and those without an internal backbone (invertebrates). Vertebrates include dogs, cats, fish, kangaroo, humans, birds etc. Invertebrates are creatures like ants, worms, jellyfish, spiders, crabs etc. Some invertebrates have exoskeletons. These are outside skeletons – something like a suit of armour (eg lobsters and crabs). They can be soft and thin or hard and tough, depending on the species of animal. There are over 40 000 different types of vertebrates and over a million types of invertebrates, so we need to break them down into even smaller groups. We divide vertebrates into five smaller groups: fish, amphibians, reptiles, birds and mammals.

One of the feature that helps us decide what group an animal belongs to is its body temperature. Birds and mammals can keep their body temperature steady. They are endothermic (sometimes called *homoeothermic*) animals. Fish, amphibians and reptiles have body temperatures that will change depending on the temperature of their surroundings. These are ectothermic (sometimes called *poikilothermic*) animals.

Animals that do not have a steady body temperature are more active in warmer temperatures. Reptiles such as snakes and lizards are often seen lying in sunshine on warm surfaces. If they become too warm they move into the shade and their body temperature will lower accordingly.

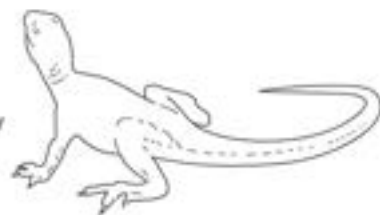
Endotherm: an animal that has a steady internal temperature due to control of metabolism

Ectotherm: an animal that has a body temperature that varies with external conditions

We sort animals into groups because of their characteristics. Think about the following five animals: galah, cockatoo, kookaburra, rosella and egret. The characteristics they have in common include being covered in feathers, laying hard-shelled eggs, having beaks etc. Because of this they are classed as being in the bird group of animals, the Class Aves (Latin, *avis* birds).

Some animals live in water and have gills to help them breathe underwater. These are fish, for example trout, barramundi and sharks.

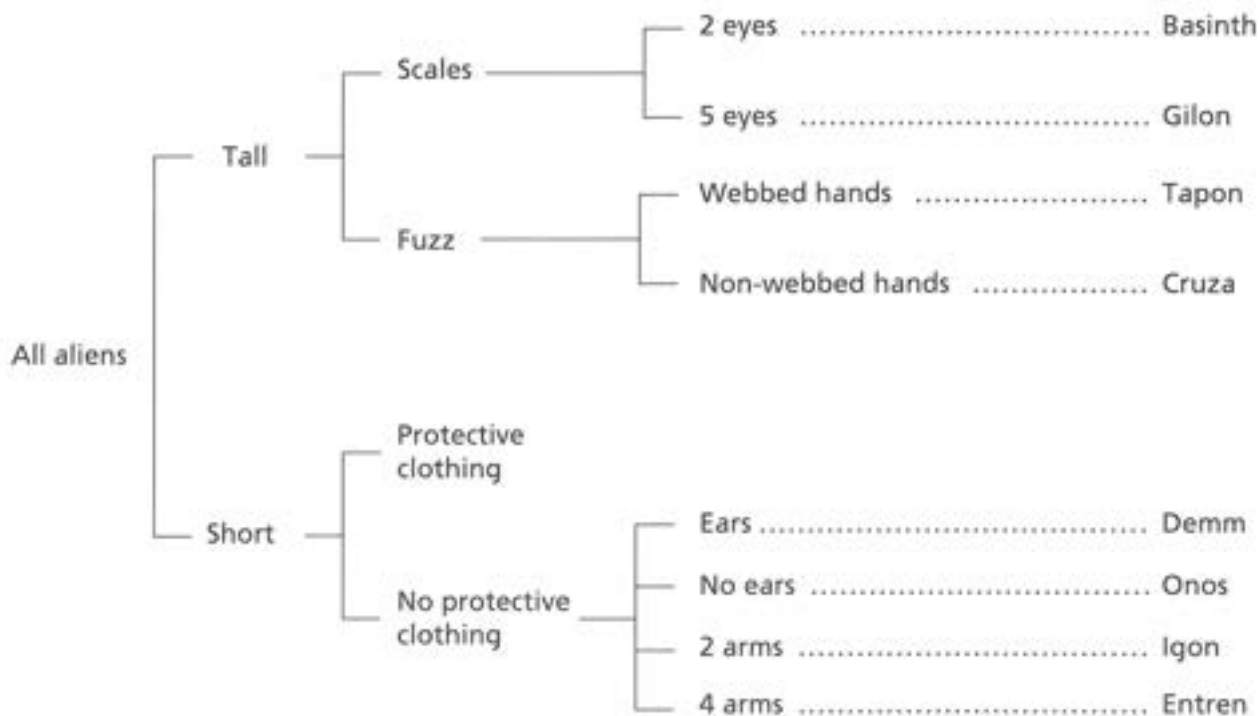
Some animals, when they are young, live in the water and have fish-like characteristics, but when they become adult change into air-breathers with lungs. These are amphibians, for example frogs, newts and toads. Amphibians lay their eggs in water and can live on land but must be near water.



- 9 Give three characteristics of birds not mentioned above.
- 10 Give three other characteristics of fish.
- 11 Species are most commonly classified according to physical appearance. Suggest three ways in which we can classify animals that do not involve physical appearance.
- 12 What is the general name for animals that cannot control their body temperature?
- 13 What is the main difference between a vertebrate and an invertebrate?
- 14 Ectothermic animals were once referred to as 'cold-blooded'. Explain why this is not an accurate term.
- 15 Vertebrates are divided into five groups. Name these groups and state which groups are endotherms and which are ectotherms.
- 16 What are the main differences between fish and amphibians?
- 17 Animals are sometimes classified according to how they move. Suggest ways in which movement can be classified.
- 18 What is an exoskeleton?
- 19 Dolphins are mammals but have many characteristics in common with fish. Name two similarities and two differences between dolphins and fish.
- 20 Show how the terms *endotherm* and *ectotherm* are derived.
- 21 The term *subspecies* is often used to refer to a population of animals. What is a subspecies?
- 22 Suggest an advantage, from the body's point of view, of being ectothermic.

Classification II: Keys

2



Consider the classification diagram above. An imaginary scientist sketched it. Read his notes below then answer the questions.

Initial observations of alien invasion

as noted by Professor Know I. Tall

Many spacecraft have landed. When the doors to these vessels opened, a variety of creatures emerged, of varying phenotypes (physical appearances). Some were tall and others short with no 'in between' sizes. Tall and short alike could be found emerging from a given ship. About half wore some form of protective clothing, others nothing. Many, not all, had webbed hands and each had either 2 or 5 eyes and 2 or 4 arms. Ears were not always present. I have drawn up a classification diagram suggesting which planets each phenotype may have originated from as based on previous research.

1. Creatures that are short, have no clothes and two arms come from which planet?
2. Creatures that are tall, have webbed hands and are fuzzy come from which planet?
3. What does the variety of phenotypes emerging from one spacecraft suggest to you? Is the Professor's suggested planet of origin necessarily correct? Why?
4. Discuss the flaws in this diagram. For example, consider where a tall creature, with webbed hands and no ears might have come from.
5. Use the Professor's 'Initial observations' notes to draw up another classification diagram using the information available. Describe how yours is an improvement on Professor Know I. Tall's.
6. Animals that live on land are *terrestrial* and those that live in water are *aquatic*. Give five characteristics that could be used to help classify aquatic animals.
7. Define *amphibian* and give two examples of an amphibian. Would amphibians be classified as terrestrial or aquatic? Explain your answer.
8. The Professor's creatures are *extraterrestrials*. What are extraterrestrials?

Classification III: Invertebrates

Invertebrates are animals without internal backbones. They are usually less complex than vertebrates. There are over a million species of invertebrates. We can subdivide invertebrates into smaller groups depending on their characteristics. The groups we use are echinoderms, coelenterates, flatworms, segmented worms, molluscs and arthropods.

Echinoderms all live in the sea. They have a spiny skin. Starfish, sea urchins, sea cucumbers are all echinoderms.

Coelenterates nearly all live in the sea. They often have stinging cells. Jellyfish, hydra and sea anemones and coral are all coelenterates.

Flatworms are very simple creatures. They live in water or as parasites (live on other creatures). They all have flat bodies. Liver flukes, tapeworms and planaria are flatworms.

Segmented worms have long, thin and rounded bodies divided into segments (parts). Earthworms and leeches are segmented worms.

Molluscs mostly live in the sea or fresh water. They usually have shells. Many have a large 'foot'. Snails, octopuses, clams and oysters are molluscs.

Arthropods all have an exoskeleton (strong outside case). This is the largest group in the animal kingdom. Arthropod bodies are in sections and their legs are in pairs and jointed. Because it is such a large group, it is further divided into insects, arachnids, myriapods and crustaceans.

Insects have six legs. Their bodies are in three parts. Many insects have wings. The bee, housefly and ant are all insects.

Arachnids have eight legs and their bodies are divided into two parts. There are over 60 000 species of arachnids. As well as spiders this class includes scorpions, mites and ticks.

Myriapods The word myriapod means 'many legs'. The two main groups of myriapods are centipedes and millipedes. They both have a body with a head and a long trunk with many segments. Centipedes have one pair of legs per segment. Millipedes have two pairs of legs per segment.

Crustaceans have gills and antennae (feelers). Many are found in the sea. Lobsters, crabs, prawns and woodlice are all crustaceans.



- 1 Name the groups that make up invertebrates.
- 2 How many species of invertebrates are there in total?
- 3 What does a parasite do?
- 4 Give an example of a parasite.
- 5 Name two examples of an echinoderm.
- 6 Name two examples of a mollusc.
- 7 Name two examples of a flatworm.
- 8 What is an exoskeleton?
- 9 How many legs do arachnids have?
- 10 How can you tell centipedes and millipedes apart?
- 11 How would you classify yabbies? Give a reason for your answer.
- 12 Draw up a table to describe the characteristics of the four sorts of arthropods (insects, arachnids, myriapods and crustaceans).
- 13 Crustaceans are sometimes called 'insects of the sea'. Do you think that this is an appropriate name? Give a reason.

Research

- 14 In the class Myriapoda, one subclass is carnivorous, while the other is herbivorous. Find out which group (centipede or millipede) is the carnivore and which is the herbivore.

Classification IV: Vertebrates

We can summarise the characteristics of the five main groups of vertebrates in a table:

Fish	Amphibians	Reptiles	Birds	Mammals
ectotherm	ectotherm	ectotherm	endotherm	endotherm
live in water	young live in water	skin covered in scales	feathers	hair or fur
have gills	adults breathe air	air breathers	air breathers	air breathers
lay multiple eggs	lay eggs without shells	lay soft-shelled eggs	lay hard-shelled eggs	live young fed on milk

1 Read the passage on fish below and give definitions of the terms in bold.

Fish are **aquatic** and are a heterogeneous group of animals – that is, not all fish fit the **criteria** used to describe fish in general. Fish have been in existence for nearly 500 million years. Their skeletons can be made of bone or **cartilage**. Most fish have a streamlined body and are propelled through the water moving their body from side to side assisted by **fins** and a tail. Fish have a heart with two chambers and obtain oxygen through **gills**. Most fish have scales (overlapping plates) covered in **mucus**. Some fish have a **gas bladder** – an organ that adjusts a fish's buoyancy to allow it to swim at different depths. Fish are ectotherms, they cannot control their internal temperature by altering their **metabolic** rate.

2 Explain the difference between ectotherms and endotherms.

3 Name three reptiles.

Reptiles (Class *Reptilia*, from the Latin, meaning 'to crawl') are covered in scales and have a three-chambered heart. All reptiles, even those that live in water, breathe through their lungs. Most are oviparous (lay eggs) but some are viviparous (give birth to live young). Reptiles control their body temperature by outside sources such as sunlight or warm ground. They are usually found in warmer regions of the Earth. In temperate regions they tend to hibernate in colder weather, and in very hot conditions they estivate (go into a torpor). Reptiles have been on Earth for around 300 million years.

Copy the following passage and use the words listed to fill in the missing words.

air, animals, backbone, clay, develop, eating, eggs, frogs, gills, moist, not, plants, shells, toads, water, heart

Amphibians are (4). Examples of amphibians are (5) and (6). Amphibians have (7) skin. Amphibians have a (8) with three chambers. Amphibians lay (9) in water. The eggs don't have (10). The young live in (11) and breathe using (12). Tadpoles eventually (13) into frogs. This change is called metamorphosis. The young tadpoles are herbivorous – they eat (14). Adult amphibians have lungs and breathe (15). They are usually carnivorous, (16) insects, worms and slugs. Amphibians do (17) have a steady body temperature. The salamander is an amphibian. Amphibians are vertebrates, meaning they have a (18). They tend to be more active at night, rather than during the (19).



20 Name three amphibians not already mentioned.

21 Would you expect to find reptiles living in or near the Arctic or Antarctic?
Explain your answer.

22 Describe what is meant by *hibernate* and *estivate*.

23 Suggest an advantage of hibernation and estivation.

24 Suggest why reptiles have not really changed over time.

Mammals: Monotremes and marsupials

Mammals are divided into three groups: *Eutherians* (eutheria), *Monotremata* (monotremes) and *Metatheria* or *Marsupialia* (marsupials). In all cases the fertilisation of eggs is internal but the subsequent path of development varies depending on the group. Monotremes are only found in Australasia. There are only three species of monotremes. These are two species of spiny anteaters (echidna) and a single species of duck-billed platypus. The platypus (found in Eastern Australia and Tasmania) was first taken to Europe as a specimen in 1798. Because it looked so strange, it was thought to be a fake made up of the body parts of several animals. Marsupials are found in the Americas and Australasia.



1 What geographical area is covered by Australasia?

Marsupials are often considered to be a peculiar group of creatures that branched off during mammalian evolution and are oddities rather than 'the norm'. The fact that they are found in limited areas of the world may encourage this idea. Some scientists would argue against this, saying that marsupials are just another type of mammal that evolved to suit its habitat.

2 Describe the main characteristic of mammals.

3 Give a definition for *habitat*.

Eutherians (eu good, best) are the largest, most dominant group of mammals and include humans. Many texts, particularly North American, will use the term *placentals* instead of eutherians. Since marsupials also possess a placenta, and could therefore be regarded as placentals, the term is incorrectly used. The placenta is an important and complex organ designed to create a temporary bond between the mother and developing embryo. It is made of a combination of the tissues of both individuals.

Placenta – from the Greek meaning 'flat surface'. The placenta is the organ in mammals that joins the uterus (womb) and foetus and carries substances between them.

4 Suggest why the placenta contains a combination of maternal and embryonic tissue. How might hormones also play an important role in the functioning of the placenta?

The placenta produces hormones and prolongs the pregnancy while delivering blood*, oxygen, nutrients and other important substances to the embryo. It also removes the embryo's waste products. After birth the placenta detaches from the mother's uterine wall and is expelled soon after the newborn is delivered. The role of the placenta differs between eutherians and marsupials.

* The human placenta is unique in permitting the transfer of blood platelets and white blood cells.

5 Suggest what triggers the placenta to end the pregnancy.

The young of the *metatheria* (meta changed), more commonly referred to as *marsupials*, also develop within the mother's body but, unlike the eutherians, they are born incompletely developed – often smaller than a sugar cube! In human terms this would be called *premature*. These animals are named after the *marsupium* (pouch) in which young complete their development. Marsupials possess a placenta but it does not produce hormones nor extend the length of the pregnancy the way the eutherian placenta does. After birth the tiny, premature young crawl into the pouch, following a path licked by its mother. The path leads to the pouch where they will attach to a teat and complete development. Marsupials are believed to be the last of the mammals to diverge from the mammalian family. *Meta* also refers to higher or transcended, as in metaphysics, which could be used to argue that marsupials are the 'higher' mammals!

6 Give a definition for *premature*.

Monotremata (mono one, trema hole) consist of the unusual, but loveable, platypus and echidna. This small group is believed to have been the first to branch off from the mammalian line during evolution. Monotremes have a single opening, the *cloaca*, through which all genital and waste matter is released, unlike the other groups that have both urogenital and anal openings. The platypus has a duck bill and webbed, duck-like feet



that possess claws and, unlike ducks, it is covered in fur.

The echidna is nocturnal and oviparous. The echidna has no teeth but a long, sticky tongue and has claws specially adapted for digging. Its spines are hair-like quills that resemble a barb more than hair or feathers. It has a gestation period of around 4 weeks. The developing embryos of monotremes do not develop internally. These animals lay eggs in which young are sustained by a nutritious yolk while they develop. The newborns are fed milk but not using a teat. The mother secretes milk onto her body and the young lick it off.

- 7 Give a definition for *nocturnal*.
- 8 Give a definition for *gestation period*.
- 9 Describe the major differences between eutherians, marsupials and monotremes.

Animals unique to Australia: marsupials

Australia is unique in the world for having the largest number of marsupial species. Marsupials are also found in some other countries but not with such diversity in species as in Australia. Marsupials are found in parts of the United States of America, New Guinea and New Zealand and a few other South Pacific Islands. They exist as carnivores and herbivores. Examples of marsupials include: Tasmanian devil, numbat, kangaroo, quoll, marsupial mole, rat kangaroo, echidna, possum, glider, dunnart.

- 10 Name two marsupials listed that are carnivores.
- 11 Name the two ant-eating marsupials listed.
- 12 The Tasmanian devil is not classified strictly as a carnivore because it will eat dead and dying flesh as well catching live prey. What is the name given to such an animal?
- 13 Classify the following animals under the correct headings below:

<i>human</i>	<i>echidna</i>	<i>Tasmanian devil</i>	<i>kangaroos & wallabies</i>	<i>rodents</i>		
<i>platypus</i>	<i>wombat</i>	<i>gliders & possums</i>	<i>apes & monkeys</i>	<i>cat</i>	<i>bilby</i>	<i>dog</i>
Eutherian		Marsupial	Monotreme			

An interesting argument proposes that marsupials may have shed the egg phase (as found in monotremes) in favour of allowing partial development within the mother's pouch. In contrast, the eutherians extended the time of development within the mother's uterus. Some scientists believe that evidence exists to suggest that marsupials and monotremes may be more closely related than was originally believed and that they share a reptilian ancestry.

- 14 Using the information available to you, how would you support the arguments for:
 - a the relationship between marsupials and monotremes; and
 - b the reptilian ancestry?
- 15 Do you support the theory that marsupials may have shed the egg phase and that eutherians extended this phase of development? Explain your answer.
- 16 The common mouse, a eutherian mammal, usually eats small insects, fruits and vegetables. The marsupial mouse also eats these things and regularly includes small vertebrates (eg lizards) in its diet. It is not a true mouse. Give the proper name for the marsupial mouse and suggest why it is called a mouse.
(Hint! see Research question below.)

Research

- 17 Give the common names for the *quoll*, *antechinus* and *dunnart*.
- 18 Based on the information provided, draw a small evolutionary tree that includes eutherians, marsupials, reptiles and monotremes.
- 19 Name the largest marsupial living today.

Species that have become extinct

The megafauna mystery

Imagine koalas the size of small bears and kangaroos twice the size of a man. Picture a leopard-sized Tasmanian devil with great shearing teeth, a 2000 kg wombat-like animal and crocodiles twice their present size. This fabulous collection of giant creatures, collectively known as *megafauna* (*mega* large, *fauna* animal), roamed Australia's woodlands and rainforests during the Pleistocene Period. Most were marsupials but an assortment of birds and reptiles also formed part of this wonderful assemblage.

1 Read the following passage and give definitions for the terms in **bold**.

Fossil evidence has revealed that the megafauna survived for thousands of years before about 70% of them began to mysteriously disappear. To date, no satisfactory explanation for the cause, or precise timing, of their disappearance has been presented. The mystery is compounded by the selectivity of the event that **eliminated** mostly large mammals and reptiles, although a range of other animals of varying sizes also vanished. Popular suggestions include Aboriginal hunting techniques or a dramatic climate change being responsible.

Many people believe that humans arrived in Australia between 40,000–60,000 years ago on rafts and dugout canoes. When they arrived the megafauna were here, apparently **thriving**. What happened to the animals after humans arrived remains a mystery. A number of scientists think that most megafaunal extinctions occurred about 25,000 years before present but not all at once or from a single cause. Others argue that the **extinctions** occurred 30,000 years ago and resulted from **overhunting**. Coupled with changes in natural bushfire patterns, new waves of extinction continued until the megafauna were completely eliminated.

Overkill theory

P S Martin, an American, wrote a paper explaining how overhunting pushed North America's giant mammoths to extinction. He argued that wherever humans settle, large animals soon disappear, citing the New Zealand *moa* as an example. Moas were between 1 and 4 metres tall and resembled ostriches, but had no wings. This led other scientists to suggest that the arrival of Aborigines in Australia had a similar effect, supporting their view with the fact that Australia also lost a great many flightless birds. However, the timing of the extinctions does not correspond with the dates Aborigines are said to have arrived. In New Zealand some flightless birds became extinct within a few hundred years of human arrival whereas dates indicate that the megafauna would have co-existed with humans for between 7000–10,000 years in Australia. If overhunting was solely responsible they would not have survived so long. Some sources point out that a climate change may have also been partially responsible for reducing the numbers of the slow-breeding moas.

2 Name a bird, not already mentioned, that resembles a moa.

3 What word in the paragraph above does this describe: '*The original (or earliest) inhabitants of a country.*'?

Another inconsistency between the North American and Australian extinctions involves the hunting tools and techniques used by Australian Aborigines. Archaeological digs have revealed that North American Indians possessed long spears and other tools suitable for killing large animals whereas the earliest Aborigines used weapons appropriate for hunting smaller game and fish. The large tools required to kill big game, such as large kangaroos, only appeared in Australia some 5000–7000 years ago, long after the megafauna were extinct. There are also many kill sites in North America, and no such sites have been found in Australia. Kill sites are mounds of bones left where great herds of animals were killed and left to rot after the carcasses were stripped. These are often found at the base of a high cliff.

4 What does a kill site at the base of a cliff suggest about how the animals were hunted? Would it be an efficient method of hunting? Discuss your answer.

5 Name two weapons that Australian Aborigines might have used to hunt wild animals.



Changes in bushfire patterns

The Aboriginal people have revealed that they once carried smouldering sticks around with them, which were used to light fires as required. Fires were lit to clear the landscape as well as to cook food. Burning the land encouraged the growth of new sprouts. The young sprouts attracted kangaroos and other grazers to the area that were then hunted. Some scientists believe that Aborigines burned the land indiscriminately, destroying many animals and their habitats and triggering a 25,000 year extinction process. Those who have taken time to understand Aboriginal land management practices discovered that in areas where traditional burning techniques occurred, there often exist more species, many endangered or rare. This suggests that the burning was not random but was done with a full understanding of its effects.

Objection to this practice arises from the fact that there were clearly two waves of extinction. The Tasmanian devil (*Sarcophilus harrisii*) and the Tasmanian tiger (*Thylacinus cynocephalus*) became extinct on the mainland at different times and long after the megafauna. If fire was the main cause they should have vanished with their giant companions.

6 What does it mean to say there were two waves of extinction?

Climate theory

A major environmental change occurred at the Pleistocene-Holocene boundary, leaving the North American mammoths extinct about 10,000–12,000 years ago and affecting other large animals around the world by about 9500 years ago. During this time Australia experienced increased wind, lower temperatures and drier weather. These changes would affect the vegetation. For herbivores with specialised diets, serious food shortages resulted. Many animals were not equipped to survive the harsher climate. Coupled with food shortages and Aboriginal hunting, a number of species may have been driven to extinction. A condition called 'lumpy jaw' has been found on animal fossils in an area called Lancefield Swamp. Lumpy jaw occurs when animals survive long periods of drought and this provides evidence that many creatures were struggling with a harsher, drier climate.

Other possible explanations

There is also a distinct possibility that some of the animals we believe disappeared in the great extinction are still with us. It is difficult to prove without any megafauna on which to perform comparative testing. Many scientists believe that the giant kangaroo and the koala could be examples, dwarfing to their present sizes in response to pressures placed on them by climate and hunting. Dwarfing does occur. Evidence has been presented to show that the European Red Deer is, in fact, a dwarfed version of an originally larger animal. During stressful times the animals best suited to survive live on to produce the next generation. If a smaller size is better suited to an environment, the larger individuals perish. Over time the smaller counterparts produce new generations of smaller offspring. Kangaroos alive today are thought to be about two-thirds the size they reached a few thousand years ago.

Select a word to match the definitions below. (Questions 7 – 12)

Mammoth marsupial smoulder drought climate dwarfing

- | | |
|---|-------|
| 7 Adjust size to become suited to conditions | _____ |
| 8 Mammal, gives birth to young that complete development in pouch | _____ |
| 9 Allow combustion, to burn without smoke | _____ |
| 10 Large extinct, elephant-like, animal | _____ |
| 11 Extended period without rain | _____ |
| 12 Pattern of weather | _____ |
| 13 Give the scientific name for the Tasmanian devil. | _____ |
| 14 Give the scientific name for the Tasmanian tiger. | _____ |

Revision assignments

The mystery of whether the megafauna vanished gradually or in one great wave remains. Were they overhunted, did changes in burning practices harm them or was climate to blame? Many believe that climate was largely responsible while others disagree. Most likely a combination of factors was responsible, with some more harmful than others. Dwarfing is an interesting, but closed, option since we are as yet unable to verify it occurred (and, if so, to which species). Perhaps many of the megafauna species that we consider to be extinct are alive and well, simply concealed in dwarfed bodies.

- 1 Write a report on, or debate, the theories presented for megafaunal extinction. Consider the pros and cons of each theory. Supporting your argument using information based on the theories, conclude with your own opinion. Put your ideas down in note form. Include a bibliography.

Mental challenge!

Research was undertaken at a number of sites where macropods (kangaroos) were often killed on busy motorways. The results indicated that in most instances, the road kills were male. Figures ranged from 65-92%, depending on location. The species varied, indicating that it is not a particular type of kangaroo that tends to be killed.

- 2 Suggest at least three possible explanations why males represented the largest proportion of kangaroos that are hit by vehicles.
- 3 What does your answer to Question 2 indicate about possible differences between male and female macropods?
- 4 There is a connection between leaving dead animals (from vehicle kills) on the road and the deaths of more, often different species. For example, leaving a dead kangaroo on a road often results in birds and lizards being killed. Why?
- 5 The cassowary, emu, ostrich, kiwi and rhea are all distantly related animals only found in the southern hemisphere. Name two things that they have in common.
- 6 Kangaroos raise many emotions among Australians. Consider the following people who might be involved with kangaroos and outline why they may be either 'for' or 'against' kangaroos.
 - a graziers
 - b conservationists
 - c motorists
 - d Aboriginal peoples
- 7 Outline the advantages of burning large areas of Australian bush.
- 8 Why might some people be concerned about current burning of bushland?



Research

- 9 Find out the role played by the volunteer organisation WIRES.

The discovery of the cell

The first cell ever observed and illustrated was the empty, dead shell of a once living plant. In 1665 the English scientist Robert Hooke (1635–1703) used a reasonably new invention, the microscope, to look closely at a thinly sliced piece of cork. The empty 'shell' of what was once living plant matter brought to mind tiny rooms called cells (from the Latin, *cellula*, meaning 'small compartment'). Some books state that Hooke named them after the cells of monasteries, while others cite the cells of honeycombs, but whatever inspired him, Hooke named them well.

It took another 170 years before the living contents of cells were discovered. Bohemian physiologist Johannes von Purkinje (1787–1869) recognised that cells had to be composed of a living matter since they formed a part of living matter.* He named the matter *protoplasm* (*proto* first, *plasm* form). French scientist Felix Dujardin (1801–1860) eventually observed the protoplasm. Dujardin called it *sarcode*, from the Greek *sarkos*, meaning 'flesh'.

* Matthias Schleiden (1804–1881) confirmed this for plants in 1838.

Not long after Dujardin observed the protoplasm, German scientist Theodor Schwann (1810–1882) made the first really detailed inspection of animal cells. It soon became clear that both plants and animals possess cells, each with structures that were unique to the group and many that were common to both. It was Schwann who showed us that every tissue of every organism is made of cells. Prior to this many people recognised that cells formed a part of an organism, but they did not realise that every structure could be reduced to cellular components. It was then understood that the husk-like walls, which formed the once living cells of the tree, had created the cork.

The scientists named above provide an excellent example of how the different branches of science can unite to advance our knowledge of a given topic. Give a brief description for the different branches of science each man specialised in.

- 1 Robert Hooke – Experimental physicist _____
- 2 Johannes von Purkinje – Physiologist _____
- 3 Theodor Schwann – Zoologist _____
- 4 Matthias Schleiden – Botanist _____
- 5 Felix Dujardin – Biologist _____
- 6 Write brief summaries about how the works of these men led to the discovery and understanding of the cell.

Research

- 7 Schwann and Purkinje had the following named in their honour: *Schwann cells*, *Purkinje cells* and *Purkinje fibres*. Investigate and briefly describe each.

The Cell Theory

When it became clear that all living things were made of cells, scientists and naturalists began taking a close look at all kinds of other things. Thanks to the microscope, tiny worlds, invisible to the naked eye, were finally revealed. One such world was the minute life forms found in pond water. Some were made of a single cell and were called *unicellular organisms* (*uni* one) while those made of many cells were named *multicellular organisms* (*multi* many).

The statement that all living things are made of cells was known as *The Cell Theory*. The original theory has been modified to state that the cell is the basic primary structure of all living things. This statement recognises that not all components of living organisms contain living cells. For example, black bread moulds and certain by-products of cellular activity do not. The newer theory also states that living cells can only arise from other living cells and that the cell is the smallest form of life.



We now accept that all living (organic) matter, from the tiny unicellular organisms to the largest multicellular organisms, is made up of basically the same components. These 'core' components are non-living (inorganic) matter. It is the arrangement of this matter that differs between organisms.

In 1855, Rudolf Virchow (1821–1902) the German anthropologist, pathologist, physician and physiologist, added to the cell theory. He came up with the statement '*Omnis cellula e cellula*', or 'all cells develop only from existing cells'. The smallest cells are those of bacteria and they range from just over 1 to about 10 micrometres (1 micrometre = 10^{-6} m = 1 millionth of a metre). The largest cells are the eggs of large birds and reptiles, which can be up nearly 10 cm in diameter.

8 Summarise *The Cell Theory* in point form.

9 *The cell is the smallest form of life.* Discuss this statement.

10 Give one example each of a plant and an animal by-product that can be described as an exception to the cell theory.

11 Give one example each of a *unicellular* organism and a *multicellular* organism.

12 Suggest why unicellular organisms cannot grow to be as large as multicellular organisms.

13 **Mental challenge!** Explain how living (organic) matter can arise from non-living (inorganic) matter.

14 Biologists often organise their study of life (and components of living things) in levels of complexity or hierarchies. Arrange the following from simplest to most complex:

cell atom organ tissue organ system multicellular organism

Explain the difference between:

15 organ and organelle

16 chlorophyll and chloroplast

17 molecule and compound

Answer true or false to statements (18–26) below.

18 Felix Dujardin was the first scientist to see *protoplasm*.

19 Cork was the first substance to reveal cells.

20 All components of living organisms contain living cells.

21 The first cell was observed in 1869.

22 Theodor Schwann made the first detailed observation of animal cells.

23 Rudolph Virchow recognised that cells arise from living cells.

24 A crocodile egg is a cell.

25 Bacteria are the smallest form of life.

26 The first cell observed was a living plant cell.

27 Suggest ways in which the scientists described in questions 1–5 might have contributed to our knowledge of the cell.

Microscopes I

9

The history of microscopy

Ever since humans have studied the world around them, people have wanted to see things with greater detail than the naked eye permitted. An important event that advanced the development of cell theory was the invention of the microscope. There is some debate about specific dates and relative importance of contributions, but the following all played a part.

In Holland, near the end of the sixteenth century, both Hans Lippershey and the father and son team of Hans and Zacharias Janssen were credited with the invention of microscopes. Dutch clockmaker Anton van Leeuwenhoek (1632–1723) was also recognised with inventing the first practical microscope-like instrument. Leeuwenhoek's device was just two lenses which, he discovered by accident, dramatically increased magnification when placed one over the other. Robert Hooke (1635–1703) built another important microscope. Hooke's reflecting microscope was described in his 1665 book *Micrographia*, along with his illustrations of many of the objects he examined using it.

A microscope with two or more lenses in it is a *compound microscope* (*compound* to combine or put together). The word *lens* comes from the Latin word for lentil, which is the shape of a convex lens.

Marcello Malpighi (1628–1694) was an Italian physiologist who, in 1661, used a microscope to study blood circulation. In 1838 Matthias Schleiden, a German botanist, studied plant parts under a microscope and discovered that plant parts are made of cells. A year later, Theodor Schwann, a German zoologist, examined animal parts under a microscope and revealed that animal parts are also made of cells.

Around 1880 to 1890 Pasteur and Koch made many discoveries using microscopes. Louis Pasteur (1822–1896) used the microscope for his work on yeast and bacteria while, in 1876, Robert Koch discovered the microbes *Bacillus anthracis* – responsible for anthrax. For this Koch was awarded the 1905 Nobel Prize for Physiology and Medicine.

- 1 What is meant when something is studied with the 'naked eye'?
- 2 Draw up a time line showing important dates in the development of the microscope and discoveries made using microscopes.
- 3 Briefly describe the difference between a simple and a compound microscope.
- 4 What disease is caused by *Bacillus anthracis*?
- 5 What is a typical size for a bacterium cell?

Research

- 6 Walther Flemming (1843–1905) used microscopes in much of his work. Outline one of his discoveries in biology. Match each branch of biology listed in the box to the correct definition below.

<i>virology</i>	<i>anatomy</i>	<i>pathology</i>	<i>mycology</i>	<i>ornithology</i>	<i>phycology</i>
	<i>cytology</i>	<i>histology</i>	<i>embryology</i>		

- 7 Study of structure of plants and animals. _____
- 8 Study of fungi. _____
- 9 Study of plant and animal diseases. _____
- 10 Study of tissues (plant and animal). _____
- 11 Study of viruses. _____
- 12 Study of algae. _____
- 13 Study of early development of an organism. _____
- 14 Study of birds. _____
- 15 Study of structure and function of cells. _____

Parts of the microscope

16 The compound light microscope used in school and university laboratories has changed little over the last 60 or so years. The boxes below show parts of a typical microscope and the functions of these parts. Cut out the boxes and match each part with its function. Stick them in your book under the headings 'Part' and 'Function'.

Ocular lens (eyepiece)	Lowest part, support for microscope
Used to get specimen approximately in focus	Has lenses of different magnification
Arm	Condenser lens
Stage	Pick up microscope by this part
Objective lens	Fine focus knob
Lens closest to specimen	Brings specimen into sharp focus
Revolving nosepiece	Microscope slide is placed on this
Mirror	Base
Reflects light onto specimen	Focuses light from mirror onto specimen
Place eye to look through this	Coarse focus knob

Microscopes II

10

Carefully read through the passage. Then select words from the list to fill in the spaces. Each word is only used once.

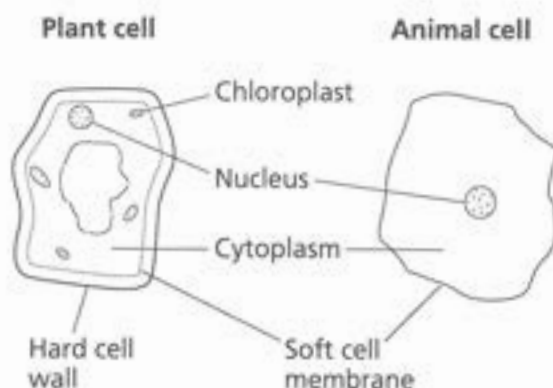
BOTANIST	CELLS	CENTURY	COMPOUND	DIVISION	DYES	ELECTRON
EYEPIECE	FOCUS	HOOKE	LENSES	LIGHT	MAGNIFIED	MICROSCOPE
MICROORGANISMS	MILLIONTHS	MOLECULES	PLANTS	OBJECTIVE	RESOLUTION	
SPECIMENS	SLIDES	SMALL	SMALLER	STAINED	THOUSANDS	VIRUSES

Microscopes give _____ images of objects that are difficult to see with unaided vision. Two main types of microscope are in common usage, the conventional _____ microscope that uses visible light and glass lenses and the _____ microscope that uses a beam of electrons. There are two types of electron microscope – the transmission _____ and the scanning electron microscope. All light microscopes nowadays are _____ microscopes, which have two sets of _____ arranged at opposite ends of a (body) tube. The _____ (ocular lens) is at the top of the tube and the _____ lens is at the bottom. The objects being looked at by microscopes are called _____ and are usually placed on glass _____. The distance from the tube to the specimen is altered to give an image that is in _____. Sometimes the parts of the specimen can't be seen clearly and need to be _____ using a dye. Light microscopes can only magnify a few _____ of times. The _____ of a light microscope is very important – this is a measure of how well a microscope can distinguish between very _____ objects that are close together. The light microscope can only give images of objects that around 500 nanometres, that is 500 _____ of a metre (10^{-9} m). The electron microscope can show items thousands of times _____ than this. Light microscopes can show the larger structures of cells and the shape of bacteria but cannot be used to observe _____. Electron microscopes can give images of viruses and even large _____.

The microscope has been an essential tool for biologists in their study of organisms and _____. The first compound microscopes were invented at the end of the sixteenth _____. The word 'cell' was first used by Robert _____ (1665). Antoni van Leeuwenhoek is credited with seeing the first living _____ (1676). Robert Brown, a _____, discovered the nucleus in plant cells (1831). Matthias Schleiden stated that all _____ were made of cells (1837) and a year or so later his colleague, Theodor Schwann, came to a similar conclusion that cells were the basic unit of animal structure. Rudolf Virchow proposed that all cells arose from cell _____ (1859). Walther Flemming pioneered the use of _____ to stain cells. He was the first to observe mitosis (1880).

Structure of plant and animal cells

Many of the features of plant and animal cells are the same as the simplified diagram indicates. Plant and animal cells (excluding bacteria) contain a nucleus (housing genetic material) as well as features that synthesise protein and energy producing components. A gel-like cytoplasm is found in both cell types and the whole cell is enclosed within a cell membrane. The membrane is a thin sheet that controls what enters and leaves the cell, using a series of pores. Plants have additional features, such as chloroplasts or other plastids (photosynthetic cells), to compensate for the fact that they must produce their own foods (via photosynthesis). Lacking a skeleton, plants need another way to maintain some rigidity, so a cell wall helps to stiffen the structure, although bark also helps with this in woody species. Most bacteria have a cell wall but no cell membrane.



- 1 Plants are *autotrophs* and animals are *heterotrophs*. Explain what these two terms mean.
- 2 What is the major constituent of cell walls in plants?
- 3 Copy the simplified diagrams of the plant cell and animal cell. On each cell draw in one more feature not already shown and label this feature.
- 4 Name the part of a cell described as: 'An organelle that controls many functions of a cell and is the site of chromosome formation'.

Eucaryotic and procaryotic cells *

Most of the cells you will encounter in biology are *eucaryotic* cells (*eu* good, *karyon* nut). Eucaryotic cells have a self-contained nucleus – the 'nut' within it – and membrane-bound organelles. Organelles are specialised parts of a cell. They include chloroplasts, Golgi bodies and mitochondria. Chloroplasts contain chlorophyll and are the site of photosynthesis. Golgi bodies are also called Golgi apparatus or just Golgi. They modify and transport proteins. Mitochondria (singular – *mitochondrion*) produce energy using cellular respiration and the chemical compound ATP (adenosine triphosphate). The genetic material of eucaryotic cells is complex, with various proteins bound to the DNA and many different chromosomes.

* Spelling variation: *eukaryotic*, *prokaryotic*.

- 5 Give the full name of ATP.
- 6 What does this sentence describe: 'Body found in the nucleus containing the genes of the organism'?

In contrast, *procaryotic* cells (*pro* before, *karyon* nut) are the simplest of cells. In these cells the organism's genetic information is not contained within a nucleus, nor does it have proteins bound to it. Bacteria are an example of these 'pre-nuclear' organisms. Bacteria generally contain a single chromosome. The membrane-enclosed organelles found in eucaryotes are absent in procaryotes. The nuclear material is not contained within a nucleus but is spread out randomly within the cytoplasm. Procaryotes are found in the kingdom *Monera*, which consists of primitive organisms.

- 7 Name two organisms found in the kingdom *Monera*.

Organelles

There is a clear division of labour among a cell's organelles. Each organelle is specialised to perform specific tasks. Organelles are suspended in a clear, colloidal, gel-like substance called *cytoplasm*. Cytoplasm is found between the outer nuclear membrane and the inner cellular membrane. (Hint: think of the children's balloons in which a smaller balloon is floating within a larger one. The cytoplasm fills the 'air space' in which the smaller balloon floats). Protoplasm consists of cytoplasm, mitochondria, nucleus and (in plants) plastids. Most organelles have a membrane (from the Latin, *membran*, meaning 'skin'), which is a thin sheet with pores (holes) covering it.

- 8 Give a definition for *organelle*.

The main components of a typical plant and animal cell are listed below (✓ = present in cell).

Organelle	Animal	Plant	Function
cytoplasm (<i>cyto</i> cell)	✓	✓	gel where organelles are suspended
nucleus ('kernel')	✓	✓	houses genetic material
nucleolus ('nut')	✓	✓	within nucleus, RNA synthesis
mitochondrion (<i>mitos</i> thread)	✓	✓	energy production
Golgi apparatus*	✓	✓	exports cellular products
starch grain		✓	important food reserve for plant
vacuole ('empty space')		✓	stores water, air and other matter
endoplasmic reticulum (ER)	✓	✓	protein modification, lipid synthesis
plastid (<i>plast</i> granule)		✓	create and store various substances
chloroplast (<i>chloros</i> green)		✓	major plastid vital to photosynthesis
cell wall		✓	protects and supports cell, rigidity
cell membrane†	✓	✓	controls what enters and leaves cell
lysosome (<i>lysis</i> destroy)	✓	✓	intracellular digestion

*sometimes called Golgi body or just golgi †also called plasma membrane

9 What is the function of the cell wall in plants?

10 Suggest why animal cells do not have cell walls.

Research

11 The plasma membrane is said to be *selectively permeable*. Define selectively permeable and give an example.

Roles of cell components

You will be familiar with the levels of organisation in a living multicellular organism. For example, cells give rise to tissue, tissue to organs, organs to organ systems and finally, organ systems to whole organisms. Each level of organisation builds on the previous one. The cell itself is also composed of specific components, called organelles (tiny 'organs'), each of which perform clearly defined tasks. Cellular components can be identified through a light microscope by fixating (killing) the cell and staining it with dyes that will make the contents easier to see. Note: some components can be identified with a light microscope, others are so small that an electron microscope is required.

12 Name three organelles that are present in plant cells but not in animal cells.

Give scientific meanings for:

13 staining 14 unicellular 15 fixating/fixation 16 multicellular 17 protoplasm

18 Suggest why some cells are examined alive and others are killed before viewing under a microscope.

19 Why is an expensive electron microscope used to study some parts of the cell, when less expensive light microscopes are available?

Cell components

The nucleus

Sometimes called the heart (or nut) within the cell, the nucleus is a self-contained unit in which DNA, the organism's genetic material, is housed. DNA, deoxyribonucleic acid, is often called the blueprint material of the body because it contains all the coding information for making molecules required by the body. DNA is a nucleic acid and the most important part of chromosomes. It is the material from which genes are formed. The nucleolus (plural nucleoli) is a spherical body within the nucleus and manufactures RNA, ribonucleic acid. RNA is another nucleic acid and assists DNA in its roles. Nucleic acids perform two major functions. They pass on hereditary information and ensure the formation of key proteins.

Because the nucleus houses the information required for normal development, it 'oversees' and instructs how development will occur.

- 1 Match the words on the left with the correct meaning on the right by joining the corresponding arrows. Then write the correct information into a table.

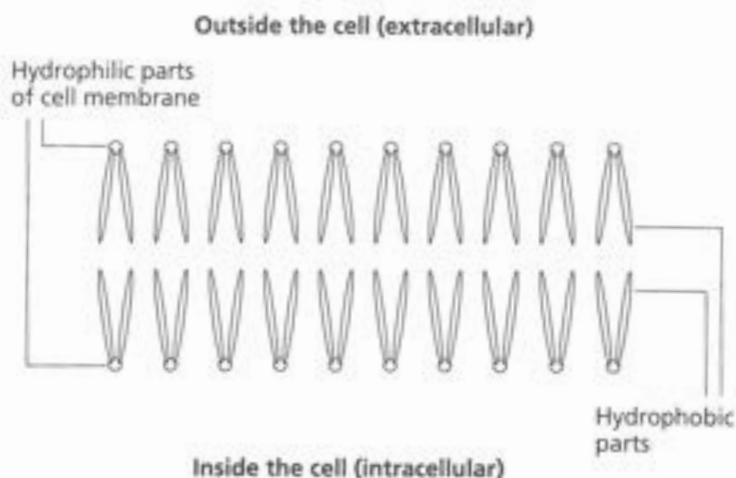
DNA	→	←	structures containing DNA
deoxyribose	→	←	unit containing a sugar, a base and a phosphate
gene	→	←	material which forms genes
chromosomes	→	←	sugar found in DNA
nucleotide	→	←	smallest unit of heredity
RNA	→	←	'little nucleus'
nucleolus	→	←	nucleic acid similar to DNA

Cytoplasm

Cytoplasm consists of all of the cell substances outside the nucleus and inside the cell membrane. It is mainly a watery fluid (cytosol) with organelles suspended in it.

Cell membrane

Cell membrane acts as an envelope in which a cell's organelles, cytoplasm and nucleus are housed. It is also the gateway to transporting materials in and out of the cell. Substances inside the cell are intracellular and those outside the cell are extracellular. The membrane is made up of a double layer of phospholipids and cholesterol, positioned in a recognisable pattern. Phospholipids are lipids (fats/oils) having phosphate groups. Cholesterol is a complex alkanol (alcohol) that exists in all animal fats and oils. The head of each phospholipid molecule is hydrophilic, or 'water-loving', while the tails are hydrophobic, 'water-fearing'. In addition to phospholipids and cholesterol, protein molecules are embedded within the structure. The membrane also has many tiny openings (pores). The exact thickness of the membrane and type of protein embedded within it varies depending on the cell.



- What are the major functions of the cell membrane?
- What is meant by *hydrophilic*?
- What is meant by *hydrophobic*?
- What is a *lipid*?
- What is a *pore*?
- Describe the differences between a cell, an organelle and cytoplasm.
- Describe the difference between the prefixes *intra* and *inter*.

9 Discuss the importance of cytoplasm.

Vacuoles

Vacuoles are found within cells. The word vacuole means 'empty space' because that is how it appears under the microscope. Generally, plant cells have large vacuoles (sometimes taking up the majority of the cell) and animal cells have a greater number of much smaller vacuoles. Vacuoles contain water and dissolved substances.

10 Would a vacuole be considered an organelle? Explain your answer.

11 Suggest the outcome of one of a cell's organelles becoming damaged or dying.

Mitochondria

Mitochondria (singular – mitochondrion, from the Greek words meaning 'thread' and 'small piece of grain') can be round or long depending on the type of cell. They are very important in providing energy for the cell using cellular respiration. Mitochondria contain fats, proteins and enzymes.

12 Give a definition for enzyme.

One example of cellular respiration is the breakdown of glucose shown by the (much simplified) equation:



Give the names of the following substances represented by chemical formulae:

13 $\text{C}_6\text{H}_{12}\text{O}_6$ _____

14 O_2 _____

15 CO_2 _____

16 H_2O _____

Endoplasmic reticulum

The endoplasm is the fluid part of the cytoplasm and a reticulum is a folded structure. The endoplasmic reticulum (ER) is a network of membranes that extends through the cytoplasm and is involved in the transport of materials within the cell. Many chemical reactions take place on the surface of the endoplasmic reticulum. Some parts are studded with ribosomes and are called the 'rough' ER. Areas not studded with ribosomes are called 'smooth' ER. *Ribosomes* are tiny granules that make proteins. Only rough ER is involved in making proteins. Smooth ER is involved in making transport vesicles.

Golgi apparatus (Golgi bodies)

This was discovered by Camillo Golgi in 1883. Golgi bodies are part of the endoplasmic reticulum and are flattened sacs. They are involved in the modification and transport of proteins and other substances.

Lysosomes

Lysosomes are small organelles containing various digestive enzymes. They break down worn-out cell parts and also any foreign materials that enter the cell.

Summarise the functions of the following organelles:

17 cell membrane

19 mitochondrion

21 Golgi apparatus

18 nucleus

20 endoplasmic reticulum

22 lysosomes

Research

23 Sometimes *lysosome* and *lysozyme* are confused. What is lysozyme?

Substances in cells I

Major chemical compounds that contribute to cell function

In order for cells to function properly, they require a regular supply of certain substances. Important chemical compounds that contribute to cell function include carbohydrates, lipids, proteins, salts, nucleic acids, water and water-containing substances. A healthy diet ensures that we receive these in the correct ratio.

Carbohydrates

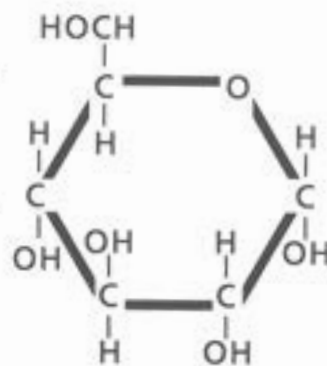
Carbohydrates (meaning *carbon plus water*) are an important source of energy and include sugars, starch and glycogen. They have the general formula CH_2O_x , or there are two hydrogen atoms present for every oxygen atom, i.e. the same ratio as there is in water. Sugars and other carbohydrates react with oxygen to produce energy. Common sugars include glucose, fructose, lactose and sucrose.

1 What do the names of the sugars have in common?

Several types of sugars exist:

- **monosaccharides**: simple sugars such as glucose and fructose (*mono* one).
- **disaccharides**: two sugars such as sucrose and lactose (*di* two). Disaccharides are made of two monosaccharides.
- **polysaccharides**: more complex sugars such as glycogen and starch. Polysaccharides are made of three or more monosaccharides (*poly* many).

Glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) is a normal constituent of the blood and tissue of animals. It consists of a single ring made up of five carbon atoms and one oxygen atom. It is a white, very soluble solid. Glucose (and other, similar sugars) reacts with Benedict's solution to change it from a blue to an orange red colour. Because glucose is so easily assimilated by the body, it is often used in 'energy' drinks and medical 'drips' where fluids are fed directly into veins.



Glucose

2 Is glucose a carbohydrate? Explain your answer.

Sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) is obtained in Australia from sugar cane and is soluble in water.

Starch is a complex carbohydrate and has the formula $(\text{C}_6\text{H}_{10}\text{O}_5)_x$, where 'x' can be from 50 to many thousands. It is a polysaccharide made up of linked glucose units and is insoluble in water. It is an important sugar storage form for plants. In animals, starch is broken down to simpler sugars in digestion. The test for starch is that it turns iodine solution from yellow/brown to blue/black.

Another polysaccharide molecule is *glycogen* (sometimes called *animal starch*). Glycogen is used to store sugars in animal cells and fungi. In animals it is stored in the liver and muscles. Animals make glycogen from glucose molecules that are not needed straight away. Glycogen is easily broken down to glucose when the organism requires a quick boost of energy.

3 Explain the difference between a monosaccharide, disaccharide and a polysaccharide.

4 Which elements are present in carbohydrates?

5 Draw the structure of a glucose molecule and give its chemical formula.

6 Why is glucose, rather than other sugars, used in 'energy drinks'?

Outline the chemical tests for:

7 glucose

8 starch

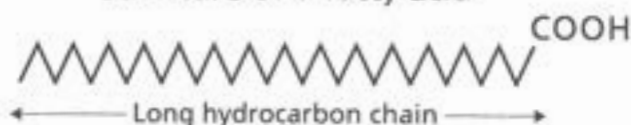
9 **Mental challenge!** Cellulose is another polysaccharide found in plants. Find out where it is stored in plants and what its main function is.

Lipids

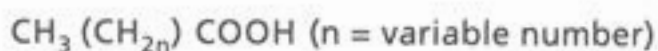
Lipids are fats or fat-like compounds (Gk. *lipos* fat) containing carbon, hydrogen and oxygen. A solid lipid is called a fat and a liquid lipid is an oil. Fats and oils are also sometimes called *triglycerides*. Lipids act as longer term, often permanent, energy storage (as opposed to glycogen, which is a more accessible 'temporary' storage system). They tend to be formed instead of glycogen when the storage space for glycogen is 'full up'. Lipids are not soluble in water. *Phospholipids* are an essential part of the cell membrane and play a part in the flow of water and water-soluble substances in and out of the cell. 'Phospho' denotes phosphorous. Phospholipids are lipids that yield phosphorous.

Fats and oils consist of *fatty acids* (types of organic acids) linked to an alcohol (alkanol), *glycerol*. Typically, fats contain around twice as much energy per gram as proteins and carbohydrates. The test for a fat or oil is that they leave a greasy mark on paper.

Structure of a fatty acid



Name the elements present in:



10 a fatty acid

11 glycerol

12 Name a health problem that could result with a diet containing too much fat?

13 What is the major physical difference between a fat and an oil?

Waxes resemble fats and oils but are harder and less greasy. When we think of waxes, beehives and candles often come to mind, but waxes have other important uses. Waxes form coatings on leaves, fruit skins, feathers and fur.

14 Suggest how wax would be useful on feathers, fur and leaves.

Steroids are also present in cells, and resemble lipids in many ways.

Proteins

Proteins (from the Greek, *proteios*, meaning 'primary') are used for cell growth and repair but can also be broken down to provide energy. Proteins are built up from amino acids. There are around 20 different amino acids and these can form many different proteins. There are approximately 30,000 different proteins present in the human body. Enzymes and most hormones are proteins.

15 Give a brief definition for *hormone*.

Plants build up amino acids from substances formed from photosynthesis. These amino acids are converted to proteins. Animals take in proteins in their diet and break these down (digest) them into amino acid components. These amino acids are carried by the bloodstream to cells, where they are reassembled into the specific proteins required by these cells.

'Biuret' reagent can be used to test for proteins. This solution contains copper ions in alkaline solution and goes purple when proteins are present. The greater the amount of protein the deeper the purple colour.

16 What is the simple test for lipids?

17 Name one element present in amino acids but not fatty acids.

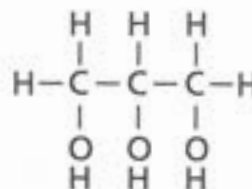
18 Are lipids soluble in water?

19 Name two uses cells make of proteins.

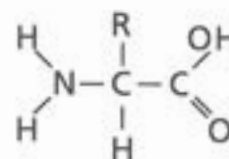
20 Outline how you could test if a substance contained proteins.

21 True or false? Proteins contain twice as much energy per gram as fats and oils.

Structure of glycerol



Structure of an amino acid



(R = various groups)

Substances in cells II

Minerals

The word 'organic' is often used to denote 'natural' or 'good for you', but when dealing with chemical substances it has a different meaning. Chemists divide compounds into two major groups, *organic* and *inorganic*. Organic compounds are those based on carbon (with the exception of carbon dioxide and carbonates). Inorganic compounds are non-carbon based. Minerals are naturally occurring inorganic substances and exist in the body as ions (electrically charged particles). Common minerals include calcium (Ca^{2+}), iodine (I^-), iron (Fe^{3+}), magnesium (Mg^{2+}), phosphorus (often in the form of phosphate, PO_4^{3-}), potassium (K^+) and sodium (Na^+) ions.

Calcium is needed in animal diets for healthy teeth and bones. Iodine is used by the thyroid gland to synthesise hormones. Iron is an essential component of haemoglobin, the red pigment in blood cells used to transport oxygen. Magnesium helps nerves cells to operate efficiently. Phosphorus plays an important part in the metabolism of cells. Potassium and sodium are involved in transferring electrical impulses in and between nerve cells.

- 1 Draw up a table giving the name, symbol and use of each mineral mentioned.
- 2 Name two food sources rich in calcium.

Electrolytes are substances that give ions in aqueous (water) solutions. It is important in blood plasma and cell fluids that there is a balance between positive ions (cations) and negative ions (anions). The major ions in blood are sodium, chloride and hydrogencarbonate (HCO_3^-) ions. If there is an imbalance in ions then health problems can result. After severe exercise (or very hot conditions) both body fluids and electrolytes have to be replaced. The level of sodium and potassium ions is of special concern to medical practitioners. Sometimes an intravenous (IV) drip is used to replace lost fluid quickly.

There is a vocabulary (often derived from Latin or Greek words) used to make up terms to describe medical conditions. Commonly used are *hypo-* (meaning lower or less than) and *hyper-* (meaning above or excessive). For example 'hyperactive' is often used to describe someone who is more active than 'normal'.

Terms relating to blood conditions often end in *-aemia* (or *-emia*). When blood is deficient in red blood cells it is called *anaemia* (literally 'without blood'). The Latin word for sodium is *natrium* and the abbreviation *nat* is often used. Similarly, potassium is designated by *kal* (from *kalium*).

Use the information above to explain what the following terms mean:

- | | |
|------------------|-----------------|
| 3 hyponatraemia | 5 hypokalaemia |
| 4 hypernatraemia | 6 hyperkalaemia |

The table below has some terms and their meanings, but not in the correct order. Copy the table and match each term with its correct meaning.



Term	Meaning
7 aqueous	a A small piece of something
8 electrolyte	b Substance that gives colour to other materials
9 fluid	c Inorganic substance found in the body
10 gland	d Substance that forms ions in solution
11 hormone	e Fluid part of blood
12 ion	f To put together
13 mineral	h Charged particle
14 particle	i Made by gland, travels in blood to affect other parts of body
15 pigment	j Substance that flows eg gas or liquid
16 plasma	k To do with water
17 synthesise	l Organ of body that makes and releases a hormone

Research

18 Describe the chemical test for the presence of the chloride ion.

Water

Since all life requires water, it is no surprise that water makes up 70-85% of a healthy cell. Water has the formula H_2O ; each water molecule consists of two hydrogen atoms and one oxygen atom. Water is required for many reactions to occur – substances in solution can move around and react with each other readily. Water bathes the components within the cell. Water (having a high specific heat capacity) helps to maintain a steady temperature for the cell, its surroundings, and the organism as a whole. Whenever you see the word *aqueous*, it means water is involved.

19 Water acts as a solvent within the body. Define *solvent* and give an example.

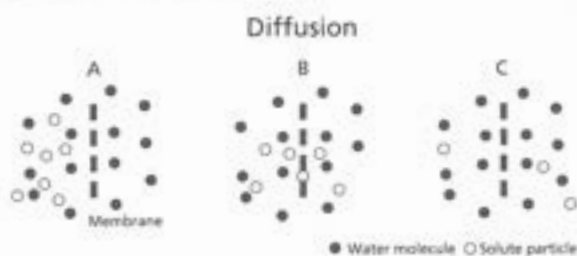
20 What is the general name given to a substance that is dissolved in a solvent?

Water helps to move substances in and out of cells by *diffusion* and *osmosis*.

Diffusion is the process where particles mix by moving randomly from a region of higher concentration to one of lower concentration to tend to equalise concentrations.

Diffusion can take place across a membrane (a thin, porous sheet). The diagram to the right shows how solute particles can diffuse across a membrane.

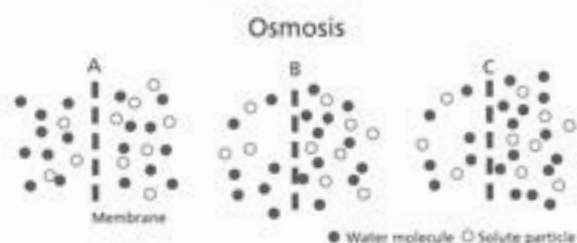
21 Copy the diagram and use it to explain how diffusion occurs.



Osmosis is the net movement of solvent particles through a semipermeable membrane into a solution of higher solute concentration that tends to equalise the concentrations of solute.

A semipermeable membrane is one that only allows certain particles to pass through. The particles will only pass through the membrane if they are smaller than the pores (openings) in the membrane.

22 Copy the diagram and use it to explain how osmosis occurs.



Both diffusion and osmosis are examples illustrating the *kinetic theory of matter*. This theory (among other things) states that particles in solution or a liquid are in constant, random motion. Another example of this is Brownian motion, named after the Botanist Robert Brown (1773 – 1858). He was studying pollen grains suspended in water and noticed that the tiny specks were moving around in an erratic manner. He initially thought (wrongly) that the motion of pollen was due to the grains themselves as they were derived from living organisms. They are being moved by the constant collisions with (invisible under the microscope) water molecules.

23 What does the word 'kinetic' mean?

24 Explain what 'random motion' means in your own words.

25 What does a botanist study?

26 What is pollen?

27 Oxygen diffuses from the alveoli in lungs to blood. Explain why this happens.

Research

28 Robert Brown had an Australian connection. What was it?

Bacteria

Bacteria can be found in almost any environment and around 2000 species have been identified. They are tiny and their sizes are usually described in μm (micrometres = 10^{-6} m).

1 What is the singular form of *bacteria*?

There are several ways to classify bacteria, one of the simplest is to categorise them by their shape:

- cocci (s. *coccus*) – spherical
- bacilli (s. *bacillus*) – rod shaped
- spirilla (s. *spirillum*) – spiral or curved.

Cocci are spherical bacteria. They can exist as single or clumped cells and cannot move themselves – they are not motile. Diplococci are paired cells (*di* two) and staphylococci form clusters (from Gk. *staphyle* bunch of grapes, which is what they resemble). Streptococci are long chains (Gk. *streptos* curved).

Identify the types of bacteria shown below, based on the descriptions given.



2 _____



3 _____



4 _____



5 _____

6 **Research:** Find a disease caused by each type of coccal bacteria.

Bacilli may possess flagella (s. *flagellum*), small hair-like appendages that move to provide locomotion for the bacteria (the name flagellum is derived from the Latin word meaning 'whip'). Spirilla (from the Latin meaning 'coil') are long curved bacteria. They sometimes have tufts of flagella.

7 What use does a bacillus make of a flagellum?

8 How many micrometres are there in a metre?

9 Cocci are described as 'not being motile'. What does this mean?

10 Identify one way of classifying bacteria.



No flagellum



Single flagellum



Many flagellum

Structure of bacteria

Bacteria are prokaryotes (variation prokaryotes); they lack a distinct nucleus. The cell is made up of a cell wall and cell membrane that encloses cytoplasm. The DNA, containing genetic information, is a single molecule in direct contact with the cytoplasm rather than being enclosed in a nucleus. Bacteria are thought to have evolved around 2 billion years ago.

11 Why might it be difficult to decide when bacteria evolved?

12 What is the shape of spirilla?

13 **Research:** Give a definition for *spirochete*.

The cell wall gives the bacteria its shape and strength. The cell (plasma) membrane is semipermeable; it controls the movement of substances in and out of the cell.

14 Define *semipermeable*.

The cytoplasm is a watery gel containing organic and inorganic substances.

Reproduction of bacteria

Bacteria reproduce asexually by binary fission. This is a process whereby an organism divides into two parts, each of which develops into two (usually genetically identical) organisms – sometimes called 'daughter' cells.

15 Outline the differences between sexual and asexual reproduction.

Binary fission

A simplified model of binary fission is shown in the diagram.
(Binary two, fission to split)

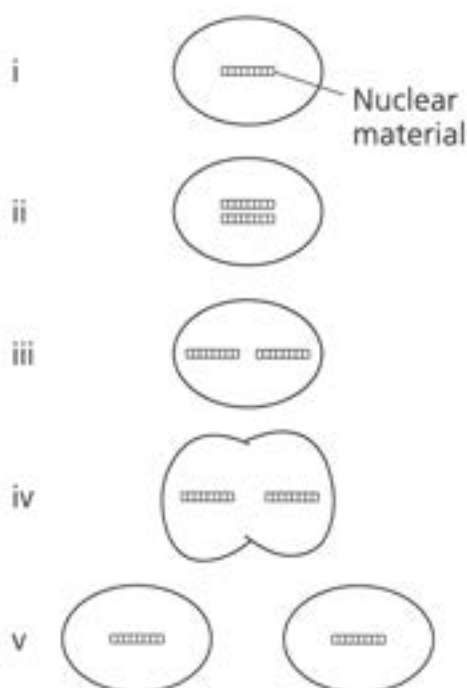
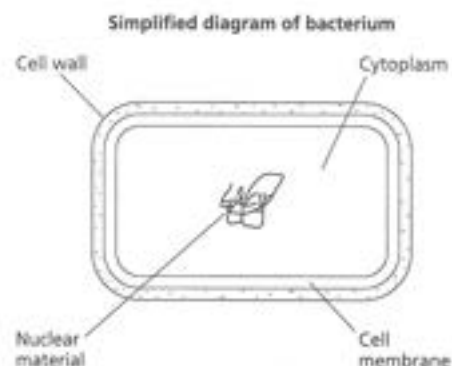
- 16 Rearrange the following statements so that they are in the right order.
- A Two daughter cells are formed, each having a nuclear area.
 - B A bacteria cell has a single chromosome made up only of DNA.
 - C The chromosomes split apart.
 - D The cell membrane moves inward to partition the cytoplasm.
 - E The chromosome is replicated.

Correct order is: _____

Compared to more complex organisms, bacteria have the potential to reproduce extremely quickly. It is not uncommon to find bacteria that, under favourable conditions, can undergo cell division (replicate) at intervals of 20 to 40 minutes.

- 17 A species of bacteria replicates every 30 minutes. Starting with a single cell, calculate how many bacteria could exist after 24 hours if this rate of replication was maintained (show working). Explain why, in practice, this number would be unlikely to be achieved.
- 18 Draw up a table that shows the major parts, and their functions, of a bacterium cell.
- 19 Read the following passage and give definitions for the terms in *italics*. You may need the help of a dictionary or text book.

Bacteria are found in practically all types of *environments*. Some types of bacteria exist in icy conditions and others near *hydrothermal vents* deep in the ocean. Some bacteria can exist in the presence of oxygen and others are *anaerobic*. Bacteria *metabolise* a wide range of materials. Gram's stain is one test used to *differentiate* between various types of bacteria. One way to classify bacteria is to group them as *symbionts* or *saprophytes*. Bacteria can affect foodstuffs in many ways – either to make them *unpalatable* or to preserve them. Some bacteria are *pathogenic* and scientists have developed *antibiotics* to use against these bacteria.



Plants: multiple choice test

- 1 Which is a major function of stems in plants?
 - A major sites of synthesis.
 - B to take water into plant.
 - C as supports for plants.
 - D to attract birds and insects for pollination.
- 2 Asexual reproduction:
 - A occurs in all plants.
 - B produces genetically diverse organisms.
 - C occurs only with human intervention.
 - D can be used to create large quantities of genetically identical plants.
- 3 Green plants can be described as autotrophic organisms. This means they:
 - A can only be grown from seeds.
 - B can be cloned.
 - C make their own food from carbon dioxide and water.
 - D head for sources of light automatically.
- 4 Some native Australian plants have special mechanisms to help them survive bushfires. Which of these can be used by a plant to regenerate after a bushfire?
 - A epicormic buds
 - B lignotubers
 - C runners
 - D waxy leaves
- 5 Abrasion, heat, smoke and leaching are all used to:
 - A preserve seeds.
 - B help some seeds germinate.
 - C kill microorganisms on seeds.
 - D sterilise seeds.
- 6 Trace elements are:
 - A required by plants in very small amounts.
 - B required by plants in large amounts.
 - C used by plants to kill parasites.
 - D made by plants.
- 7 Some Australian plants have become endangered because of:
 - A introduced plants and animals.
 - B land being cleared for agriculture.
 - C pesticides and fertilisers entering waterways.
 - D all of the above.
- 8 Which is a strategy used to conserve Australian plant species?
 - A ensure regular bushfires.
 - B eliminate introduced species.
 - C irrigate areas containing native plants.
 - D put fertiliser on areas containing native plants.
- 9 The different species living in a particular environment is called:
 - A a population.
 - B biodiversity.
 - C ecology.
 - D the gene pool.
- 10 Which of the following inhibits germination?
 - A heat beds
 - B lime pelleting
 - C cold storage
 - D misting
- 11 'A tree or shrub that grows in shallow and muddy, salty water, especially in tidal estuaries' is most likely to be a:
 - A corm.
 - B mangrove.
 - C epiphyte.
 - D Wollemi pine.
- 12 'Tissue culture':
 - A grows large numbers of genetically identical plants.
 - B grows seedlings on moist tissue paper.
 - C repairs damaged tissues in plants.
 - D is only effective with native Australian plants.
- 13 Which of the following sensors would be best suited to a computer controlled watering system? One that measures:
 - A humidity.
 - B pH.
 - C temperature.
 - D light intensity.
- 14 Overwatering of plants can:
 - A cause them to grow too quickly.
 - B cause them to develop deep roots.
 - C cause salt build up in upper layers of soil.
 - D cause them to photosynthesise too much.
- 15 It is essential for plant species to have genetic diversity:
 - A in order to look different.
 - B so they are not eaten by the same animals.
 - C so they can reproduce effectively.
 - D so they can deal with changes or diseases.

Life is a complicated process and many conditions are needed for life to continue.

1 Give five typical features of a living organism.

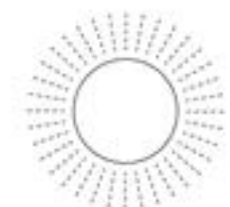
Different types of organisms require different conditions, but all need energy and raw materials to replace cells and build new ones. All living things have a *metabolism* (Gk. *metabol* change). Metabolism is the sum of the chemical reactions taking place in an organism. These reactions take in new materials and generate energy. Plants and animals have their own specific requirements, but both need energy and raw materials. Green plants are *autotrophs*; they can produce their own food by photosynthesis and generate energy through respiration.

Photosynthesis and respiration

Green plants use water and carbon dioxide (and enzymes) to produce their own food – known to us as carbohydrates. Light energy drives these reactions.

Photosynthesis occurs when plants convert radiant energy into sugars, water and other forms of energy, which are then used by the plant for growth and to maintain its health. Plants contain specialised organelles called *chloroplasts* which are not found in animal cells. Only chloroplast-containing cells can photosynthesise. The chloroplast contains several pigments that trap light. The pigment chlorophyll absorbs the red and blue light rays that make up white light. There are several types of pigment in plants and four types of chlorophyll. When a plant is exposed to radiant energy (sunshine), the light rays cause a movement of electrons within the atoms of the leaf.

This, in turn, activates the ATP/ADP. During this reaction energy is released that allows for the photosynthesis reaction to occur.



Photosynthesis
and respiration



Respiration

carbon dioxide + water + radiant energy + chlorophyll produce carbohydrates and oxygen.

Photosynthesis produces glucose as shown by the equation:



All cells need a constant supply of energy and the energy is produced by respiration. Respiration involves many stages but can be summarised thus: $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{H}_2\text{O} + 6\text{CO}_2 + \text{energy}$.

Again, enzymes are an essential part of the reactions. Note: the energy above is equal to 2881.2 kJ.

2 Summarise, in words, the photosynthesis and respiration equations.

3 Discuss the statement: 'Photosynthesis only occurs when sunlight is shining on a green plant and respiration only occurs when a plant is in the dark.'

4 Name another nutrient (not already mentioned) needed by plants.

Humans, like all animals, are *heterotrophs* – meaning they cannot manufacture their own food. Heterotrophs need complex organic compounds. The food humans eat must contain enough water, energy, vitamins, minerals and body-building substances to keep them healthy. Humans need a balanced diet, that is one that has all the substances we need and in all the right amounts.

Although water contains no energy and is not really a type of food, it is essential for our health. Water makes up approximately 70% of the body. Every day some water is lost through the kidneys, sweating and breathing out (exhaling). These losses must be replaced otherwise dehydration will occur. We could live for several weeks without food but only a few days without water. Some foods, especially vegetables, contain a large amount of water.

- 5 Outline the major difference between an autotroph and a heterotroph.
- 6 Discuss the statement: 'Respiration is the reverse chemical reaction to photosynthesis'.
- 7 What does this statement define: 'A process involving an unusual loss of body fluids'?
- 8 What does this describe: 'Originally called vital amines, these naturally occurring substances are needed in very small amounts for the regulation of metabolic processes.'?
- 9 Why is water not considered a food?
- 10 Name two ways in which the body loses water.
- 11 If water is not considered a food, why is it so important to our health?

Indicate whether statements 12 to 16 below are true or false.

- 12 Some animals drink very little; they gain most of their water requirement from food.
- 13 Chlorophyll is a green substance in plants needed for photosynthesis.
- 14 People eating a balanced diet need to take extra vitamins to keep healthy.
- 15 Glucose is a type of mineral.
- 16 Enzymes are biological catalysts.

Energy is measured in *joules (J)*, named in honour of British physicist James Joule (1818–1899). A joule is a very small unit of energy, so we tend to use *kilojoules (kJ)* instead ($1000\text{ J} = 1\text{ kJ}$). Energy is needed for all living processes, which is why our body demands a continuous supply of food. During sleep the tissues of the heart, lungs, kidneys, liver and other organs continue to work. Even the brain and muscles, although relatively inactive, use a quantity of energy. The energy we use depends on what we are doing, and can vary considerably from person to person, but the following table shows some typical figures in kilojoules per hour.

Activity	kJ used per hour
sleeping	65
light housework	600
jogging	400
squash	900
writing at a desk	160
road digging	1300
tennis	700

- 17 What is the total energy used by a student carrying out the following activities?
(Show working.)
 - a Jogs 15 minutes to the shops to buy a paper and jogs back.
 - b Spends 1 hour tidying up room, followed by 1 hour 30 minutes doing homework.
 - c Tired by all this, a nap of 30 minutes.
- 18 A person who leads an inactive life uses less energy than a very active person. If we take in less energy through food than we use up, we get thinner. What happens if we take in more energy than our body uses?

Growth also requires energy. Adolescence is one example of this. Adolescence usually happens between the ages of 11 to 18 years and frequently growth occurs in spurts rather than in a regular manner. Adolescents need as much (or more) energy from foods as adults do.



Carbohydrates are energy foods. The most common are starches and sugars. Plants, and food obtained from plants, are often high in carbohydrates. If we add a solution of iodine to a food containing starch it turns to a blue-black colour. Foods high in starch include barley, rice, wheat and potatoes. Different foods have different energy values. Sugars include sucrose (table sugar) and glucose. Glucose is very quickly absorbed into the bloodstream, which is why it is often given to invalids or patients recovering in hospital. (Whenever you see a food or ingredient that ends in '-ose', then it is a type of sugar.) Some sugars (eg glucose) can be detected by heating with Benedict's reagent, which will go from blue to brick-red.

19 Name a food that is rich in carbohydrates.

20 Suggest how giving glucose to recovering patients might be of benefit to them.

Fats and oils are even more concentrated energy foods than carbohydrates. They take longer to digest than carbohydrates. Fats tend to come from animal sources – meat, butter, eggs etc – but can also be obtained from plants, eg olive oil, sunflower oil etc. We need a certain amount of fat in our body, to store energy and act as insulation. Fats and oils give a pink colour when heated with a chemical called Sudan III and also leave a greasy spot when placed on paper and allowed to dry.

21 Name another food group apart from those already mentioned.

22 What is one way the body makes use of the food group you gave in question 21?

23 You think that a food contains starch and/or glucose. How would you test for these?

24 What type of food is *lactose*?

25 What is the major difference between a fat and an oil?

26 Why do we need a supply of fat in our body?

27 After exercise, athletes might take a drink containing glucose rather than 'ordinary' sugar (sucrose). Why might they do this?

28 In a hospital, plants and flowers are often given to patients. Why might these plants and flowers be removed from the patients' bedsides at night?

29 Why does the body use energy even when we are asleep?

30 Why can adolescents require as much food as an adult?

31 Aside from keeping our weight down, what are three other ways in which the body benefits from regular exercise?

32 There are many diets that claim to make people lose weight fast by following a special regime. Some advocate eating mainly protein and very little carbohydrate. Discuss why you should be careful before undertaking these (or any other) diets.

33 Many people follow a vegetarian diet. Outline the cases for and against being vegetarian.

34 What is the difference between a vegetarian and a vegan?

Give definitions for the following terms. You may need to use a text book or dictionary.

35 adipose tissue

36 obesity

37 nutritionist

All diets need a certain amount of cholesterol, which is used in making cell walls. However, some foods are very high in cholesterol. High levels of cholesterol in humans can cause health problems.

38 Name two foods that contain large amounts of cholesterol.

39 Outline some health problems associated with high levels of blood cholesterol.

40 Identify three ways of preserving food.

Ectotherms and endotherms

Multicellular animals can only survive within a relatively narrow band of temperatures. There are exceptions but the lower limit is usually 0°C (freezing point of water) and 45°C (when proteins are *denatured*, or 'cooked'). The optimum temperatures for particular organisms are even more restricted.

1 Explain what is meant by *optimum temperature*.

Cellular respiration releases energy when glucose is oxidised (combines with oxygen to form carbon dioxide and water). Most of this energy is released as heat and will affect the body temperature of the organism. We can divide animals into two groups, depending on how their body temperature is regulated – *ectotherms* and *endotherms*.

Ectotherms

Many animals lose most of the heat energy generated from cellular respiration to their environment. Their body temperature is variable and is determined by their surroundings. These animals are ectotherms. You may also see such animals referred to as being *poikilothermic* (having variable temperatures) or even (inaccurately) as being *cold-blooded*. Usually their body temperature is only slightly higher than ambient temperature but, depending on their activity level, can be much warmer than their surroundings.

2 Explain what is meant by *ambient temperature*.

Animals can absorb heat from the sun or hot surfaces (generally heated by the sun). They lose heat by radiation from their bodies, evaporation (perspiration etc) and conduction to cold surfaces. Water conducts heat better than air does, but large masses of water vary less in temperature than the atmosphere. This is because water has a greater heat capacity than air and takes longer to heat up and cool down. *Aquatic* ectotherms are well adapted to the temperature of their surroundings. *Terrestrial* ectotherms need to have greater responses to extremes of temperature.

3 Distinguish between the terms *aquatic* and *terrestrial*.

4 Suggest why terrestrial ectotherms must have greater responses to temperature extremes than aquatic ectotherms.

The metabolism of an organism is the sum of all of the chemical reactions occurring in that organism.

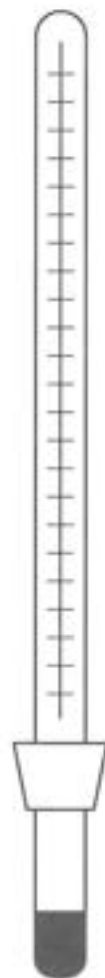
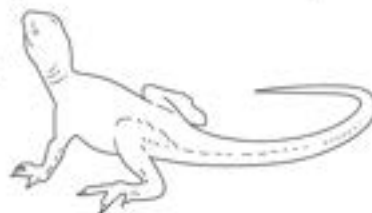
The activity of ectotherms varies with temperature. The rate of their metabolism increases when temperatures rise. If temperatures are low, the rate of reactions (both those that break and make molecules) are slow and ectotherms are sluggish, ie move and respond slowly. When land-based ectotherms need to raise their temperature they expose more of their body to the sun ('sunbaking'). Lizards and butterflies commonly do this.

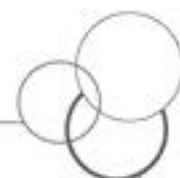
5 Name two Australian ectotherms that might sunbake.

If there are prolonged periods of cold weather then ectotherms can slow their metabolisms down considerably and become inactive. When reptiles do this they are said to be *torpid*. Insects have a similar process called *diapause*.

6 Name an Australian reptile that might become torpid.

7 Name an Australian insect that might go through diapause.





The table below shows the relative metabolic rates at different temperatures for an organism.

Temperature (°C)	0	10	20	30	40
Metabolic rate (arbitrary units)	1	2	4	8	16

- 8 Draw a best fit line graph using these figures. (Draw temperature on x-axis.)
- 9 Describe the relationship between metabolic rates and temperature.
- 10 Explain why the temperature doesn't go past 40°C in this table.
- 11 Give an explanation as to why insects in alpine areas, as a rule, tend to be small and dark.

Endotherms

Some animals (mainly birds and mammals) have mechanisms that cut down loss of heat energy to surroundings. They can use the energy generated by cell respiration to keep a relatively steady body temperature. They are also called *homoeothermic* (uniform temperature) or (inaccurately) *warm-blooded*.

- 12 Birds have feathers as insulation to minimise heat loss. Name two other specific animal adaptations that reduce heat loss.

When conditions become too cold, some endotherms hibernate. The animal seeks out a sheltered place. Its metabolism slows down and body temperature drops. The mountain pygmy-possum hibernates below a layer of insulating snow. Some frogs and lizards (which are ectotherms) use a similar strategy.

The metabolic rate of endotherms is higher than that of ectotherms and they remain active over a wider range of temperatures. The higher rate of metabolism means that endotherms use more energy and hence need to consume more food than an ectotherm of equivalent size.

- 13 Outline why it might be an advantage for an endotherm to be more active than an ectotherm.

Endotherms that live in hot areas, for example deserts, have techniques to keep cool. They can burrow underground, become nocturnal or minimise their activity during hot spells.

- 14 Define *nocturnal*.

Evaporation of water from bodies by sweating or panting can be used to keep animals cool. This can cause problems for animals living in hot areas where water supplies are scarce. Blood is an important way of moving heat around the body. When heat needs to be radiated from the body, blood flow near the surface is increased. When conditions are cold and heat loss needs to be minimised, blood flow near the surface of the skin is reduced.

- 15 Explain why a red kangaroo might lick the inside of its paws when temperatures are high.
- 16 What is the advantage for an animal such a rabbit-eared bandicoot to have large ears?
- 17 Divide the following Australian animals into ectotherms and endotherms:

barramundi bats blue-tongued lizard Bogong moth echidna
emu frill-necked lizard goanna green tree frog
platypus shark sugar glider white tip spider



- 
- 18 Endotherms (if sufficiently insulated) can cope better with conditions much colder than their body temperature than they can with conditions above their body temperature. Give a possible reason for this.
- 19 Bees are ectotherms. In the winter, if the temperature in a hive falls below a certain point, the bees become very active. In the summer, if the hive becomes too warm the bees beat their wings much more vigorously than normal. Give a possible reason for each of these two occurrences.

What do the following statements define?

- 20 'A state in which the body temperature, heart rate and respiration of an animal fall well below normal'.
- 21 'All of the chemical processes taking part in the body, including those that break down large molecules and those that make large molecules'.

Decide whether the following statements (22–31) are true or false:

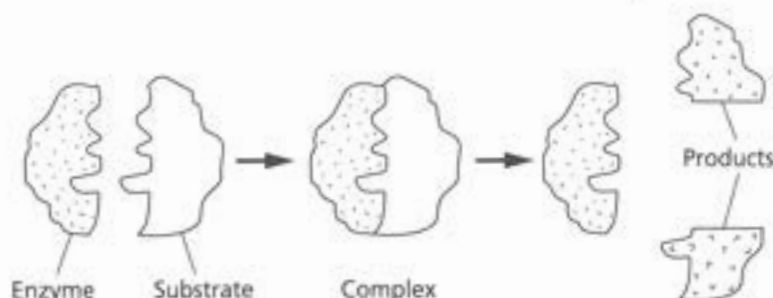
- 22 Aquatic ectotherms must be able to respond to greater temperature extremes _____
than terrestrial ectotherms.
- 23 Ectotherms obtain most of their body heat from their surroundings. _____
- 24 Invertebrates and fish, reptiles and amphibians are endothermic. _____
- 25 *Torpid* and *diapause* refer to periods of increased activity. _____
- 26 CO_2 and H_2O are products of cell respiration. _____
- 27 Multicellular animals can only survive within a narrow temperature range. _____
- 28 Very little of the energy produced during respiration is released as heat. _____
- 29 Ectotherms are also called poikilotherms. _____
- 30 Hibernation can occur within both endothermic and ectothermic species. _____
- 31 Endotherms obtain most of their body heat from their cell metabolism. _____
- 32 Define *hibernate* and give some of its benefits.
- 33 Animals that live in cold climates (eg the arctic) often have a layer of insulating fat (adipose tissue).
Name two examples of these types of animals.
- 34 Describe how humans can deal with hot conditions using technology.
- 35 Bees live in hives. Identify another type of insect that lives in similar conditions.

Homeostasis is the maintenance of a balance between the functions carried out inside an organism and the environment, including body temperature and regulating levels of chemical compounds. Animals can be divided into *endotherms* or *ectotherms* depending on how their body temperature is regulated.

1 Outline the major differences in how the body temperatures of *endotherms* and *ectotherms* are regulated.

There are many factors that play a part in an animal's metabolism. Many complex reactions are occurring, both making large molecules and breaking them down. *Enzymes* (biological catalysts) are a vital part in ensuring that these reactions occur efficiently and at body temperature. Enzymes are very specific catalysts – most enzymes only catalyse a specific biochemical reaction. Enzymes are proteins that combine with a reacting molecule, the *substrate*. A simplified version of one model (*lock and key*) is shown in figure 1.

Figure 1



2 Copy figure 1 and use it to describe how the 'lock and key' model explains how enzymes work.

The shape of the enzyme might not be quite as fixed as suggested by this model. An *induced-fit* model may be more accurate. Some enzyme molecules may have their shape changed slightly by the substrate in order for the substrate to fit into the active site of the enzyme.

3 Using your diagram of figure 1, explain what is meant by the active site of the enzyme.

Reactions that give out heat are *exothermic* and the energy levels of products are lower than that of reactants. All reactions need *activation energy* to get them started. Catalysts lower the activation energy for reactions and enable them occur more quickly. Figure 2 shows the energy changes involved when a reaction is carried out both with and without a catalyst.

4 Copy figure 2. Identify points A, B, C and D on figure 2, choosing from: Activation energy (catalysed reaction), Activation energy (uncatalysed reaction), Energy, Energy released by reaction.

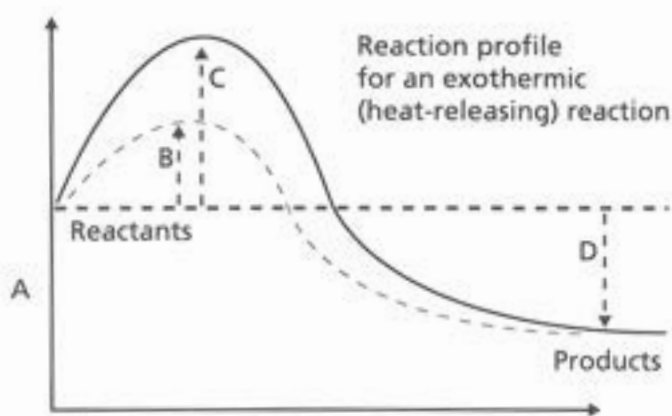


Figure 2

5 What does this define: 'A complex protein made by a living cell that catalyses a specific chemical reaction at a lower temperature than might normally be possible'?

State whether the following statements are true or false:

6 All enzymes have the ending '-ase'.

7 Enzymes are usually small molecules.

8 Enzymes are specific in their actions.

9 In a reaction you start with products and end up with reactants.

10 Exothermic reactions take in heat.

11 Enzymes are proteins.

Enzymes

Enzymes catalyse almost all of the reactions taking place in organisms and can be affected by:

- temperature
- pH
- substrate

Temperature

Figure 3 shows how the rate of an enzyme-catalysed reaction varies with temperature.

- 12 Copy figure 3 and describe how the rate of the reaction shown alters with temperature.

The *optimum temperature* for most enzymes is the temperature at which they are most efficient. In most mammals this is around body temperature. If the temperature rises much above 40°C, the enzymes are *denatured* – their structure is changed.

- 13 State the 'normal' internal temperature of a human.

pH

This is a logarithmic scale that indicates if a substance is an acid, base or neutral. Enzymes have an optimum pH range that varies from enzyme to enzyme. Enzyme activity decreases markedly more than one pH unit below or above optimum pH. Enzymes that work in the stomach will have an optimum pH of around 2, while those in human blood will have an optimum pH of just over 7.

- 14 A particular enzyme has an optimum pH of 6.5. Draw a labelled diagram (similar to figure 3) that shows how rate of reaction compares with pH for that enzyme.

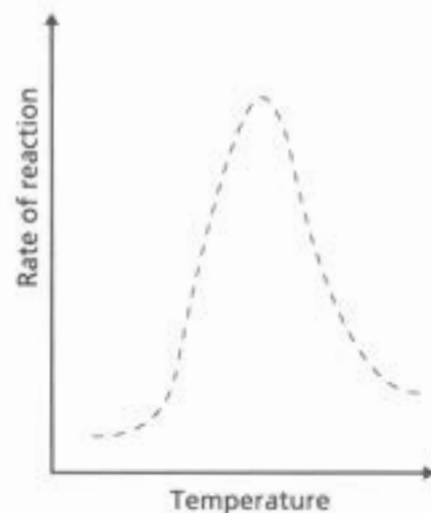


Figure 3



Figure 4

Substrate concentration

Figure 5 shows how altering the amount of substrate present (keeping enzyme concentration constant) affects how quickly a particular reaction occurs.

- 15 Copy figure 5 and describe how the rate of the reaction shown alters with substrate concentration.

Identify the terms described by the following definitions:

- 16 'The substance in a reaction acted on by an enzyme'.
 17 'A number that indicates whether a substance is acidic, basic or neutral'.
 18 'Process that changes structure of enzyme (eg by heating) so that active site no longer functions'.

State whether the following statements are true or false:

- 19 Enzymes allow reactions to take place at moderate temperatures.
 20 Enzymes are made in cells.
 21 Breaking down large molecules is called 'synthesis'.
 22 Enzymes involved in digestion in the stomach work best at pH values over 7.

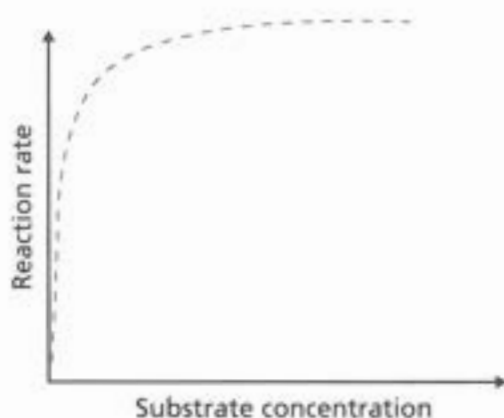


Figure 5

Human systems I

Humans (and other animals) have specialised structures to obtain nutrients from their environment. These structures are made up of many components.

1 Arrange the following in terms of increasing complexity:

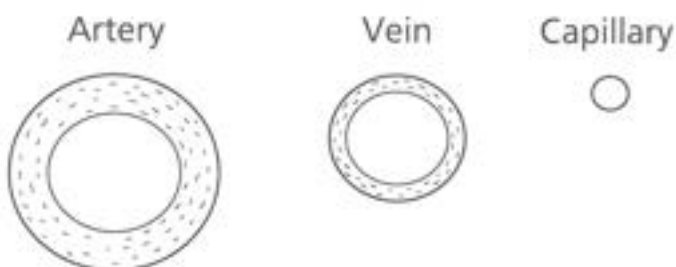
organs, atoms, cells, tissues, organism, molecule, organ system

Human systems involved in the maintenance of life include those dealing with circulation, respiration, excretion, nerves and glands.

The circulatory system

The circulatory (or vascular) system is a closed network of blood-filled tubes, or blood vessels. Blood is kept flowing one way through these vessels by the pumping of the heart. The heart is a muscle with four compartments. The upper two compartments are called *atria* and the lower compartments are *ventricles*. An adult human has about 5.5 litres of blood in circulation. Blood distributes heat and many important materials throughout the body. How often a heart beats (contracts) is called the *pulse rate*.

Cross section of blood vessels



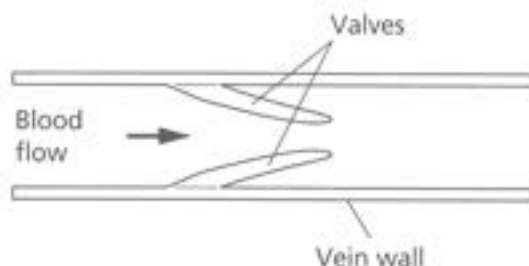
There are three main types of blood vessels: *arteries*, *veins* and *capillaries*.

2 Study the diagrams of blood vessels. Compare the relative quantity of blood each is likely to be able to carry. Explain how you reached your conclusion.

Arteries are wide, thick-walled vessels which carry blood away from the heart. Arterioles are smaller arteries that branch from larger arteries. All arteries (apart from those going to the lungs) contain dissolved oxygen and are bright red in colour. All arteries also carry blood that contains dissolved food and waste products. The food reaches body cells by means of the capillaries. Non-gas waste products are taken to the kidneys where they are processed and eventually passed out of the body as urine.

Capillaries are narrow, thin-walled vessels between 0.13 mm to 0.20 mm in diameter. They branch off from arterioles and then rejoin to form veins. Capillaries connected to the digestive system pick up dissolved food. Those connected to the lungs deliver carbon dioxide to the lungs and take away oxygen. Capillaries allow food, water and dissolved oxygen to pass out of them to cells. Carbon dioxide and other waste products pass into capillaries from cells. The very thin tissue layer (endothelium) of the capillary walls allows easy passage of substances through the capillary walls. Capillaries don't actually touch individual cells but reach cells by means of tissue fluid, a substance that links the circulatory system and the lymphatic system (the body's 'drainage' system).

Veins are, in structure, similar to arteries, but carry blood to the heart. They contain valves to stop blood from flowing backwards. These valves are flaps of tissue that allow the blood to flow one way. A forward flow pushes the flaps open and when the heart relaxes, blood in veins tends to flow back. The flaps are pushed together and stop any reverse flow.



3 Draw up a table that summarises the roles and properties of human blood vessels.

4 What role do valves play in veins?

5 In your own words, explain how valves in veins work.

6 Identify another part of the circulatory system, apart from arteries, that has valves.

7 Name two roles played by blood.

8 How many compartments are there in the human heart?

9 Explain why arteries have thicker walls than veins.

Research

10 Outline the role of the lymphatic system in the human body.

11 Outline the role played by William Harvey (a seventeenth century scientist) in the discovery of blood circulation.

The components of blood

Blood is a liquid, containing many substances, that flows through the body. Blood is moved mainly by the muscular contractions of the heart, but the blood vessels and lungs also play a part. Blood also distributes heat around the body, takes nutrients and oxygen to cells and takes waste away from them.

12 Name a waste product that is taken away from cells.

The temperature of blood is usually approximately 37.4°C in a healthy individual and has a pH of close to 7.4 – which is slightly alkaline. Blood sugar concentration tends to be in the region of 0.1%.

The four main components of blood are *plasma*, *red blood cells*, *white blood cells* and *platelets*.

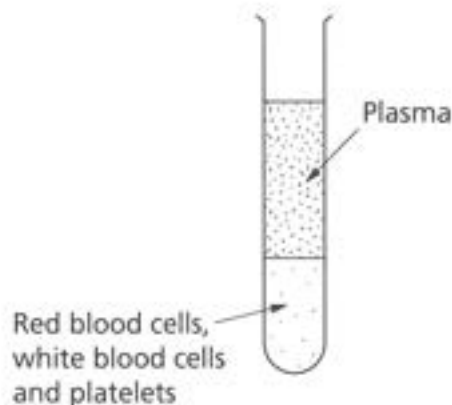
Plasma is a pale, yellowish liquid that is around 90% water. If blood is treated to stop it clotting and allowed to stand for a while, blood cells and platelets settle, leaving plasma as a liquid on top.

Plasma contains dissolved food to take to cells and waste materials being removed from cells. Also present are antibodies, enzymes and various hormones.

Red blood cells (also called red corpuscles or erythrocytes) are made in the bone marrow. They are the most common type of blood cell and are disc-shaped and do not contain a nucleus. They contain *haemoglobin* and carry oxygen around the body. Red blood cells are concave and have a diameter of around 0.008 mm.

White blood cells (also called white corpuscles or leukocytes) are larger than red blood cells. They are opaque and have large irregularly shaped nuclei. White blood cells play an important role in the body's defences against illness; they make *antibodies* and can attack bacteria.

Platelets are minute, flattened bodies made of protoplasm and are important in the clotting of blood.



13 In a table form, outline the properties and uses of the four major components of blood.

Identify the terms described by the following definitions.

14 'Substance produced by white blood cells which attacks foreign objects in the blood stream.'

15 'Clear, watery solution making up around 55% of volume of blood.'

16 'Cells lacking a nucleus which only live for about 90 days. They transport oxygen in blood.'

17 'Protein that acts as a catalyst in biochemical reactions.'

18 'Substance produced by a gland that travels in bloodstream and affects body.'

19 'Part of the cell that controls activities of the cell.'

20 'Substance through which light does not pass.'

There is a network of capillaries in the dermis (outer part of the skin). If the body becomes too warm, through exercise or climate, then the capillaries dilate (widen). This allows more blood to flow near the surface of the skin and heat is lost to the atmosphere. If the body is too cold then the rate at which heat is radiated needs to be reduced. To do this blood vessels near the surface of the skin constrict (narrow) and reduce the flow of blood.

21 What is the 'normal' internal body temperature of a human?

22 What is the pH of blood?

23 Outline how blood helps maintain a steady temperature in the body.

24 Arrange the following blood vessels in the order they receive blood from the heart: vein, artery, capillary.

Research

25 Plasma and other components of blood can survive being separated by centrifuging. Briefly describe how this process is carried out.

Multiple choice quiz

Choose the best response for each question.

26 Which statement about the heart is *not* true?

- A It is a specialised muscle.
- B It consists of four major chambers.
- C It has valves.
- D It can be consciously controlled by our brain.

27 Which is *not* a blood vessel?

- A vein
- B artery
- C lymph
- D capillary

28 Which of the following is used to listen to a heart beat?

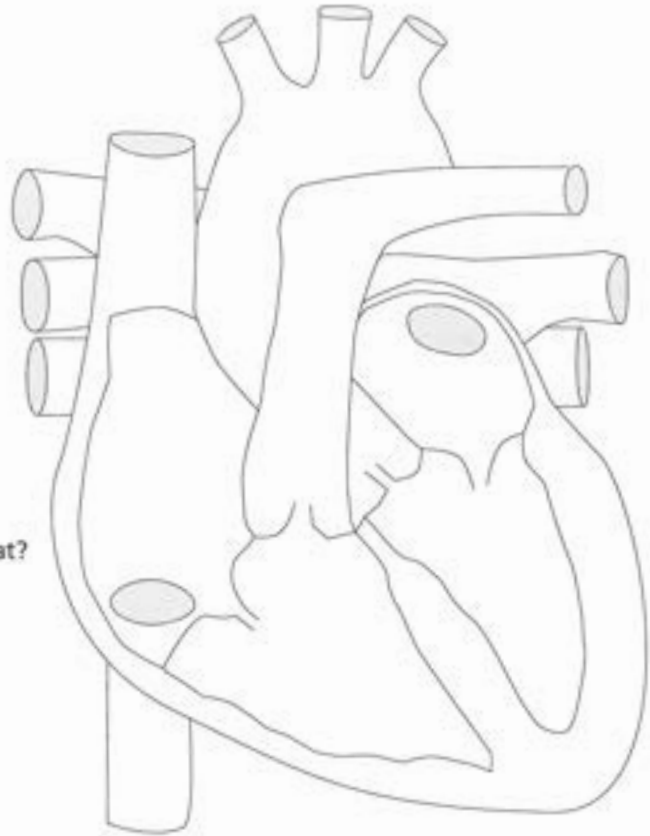
- A stethoscope
- B sphygmomanometer
- C electrocardiogram
- D endoscope

29 Blood in arteries:

- A travels away from the heart.
- B travels to the heart.
- C remains stationary.
- D doesn't have a pulse.

30 Blood is prevented from travelling in the 'wrong' direction by:

- A the heart beating.
- B valves.
- C capillaries.
- D glands.



Human systems II

Respiration

Respiration is a process by which cells generate energy.

Respiration is the breakdown of large molecules into smaller ones with the release of energy.

Aerobic respiration has oxygen as one of the reactants and *anaerobic* respiration doesn't involve oxygen. This section only discusses aerobic respiration.

Respiration involves the following processes:

- getting oxygen to the cell.
- chemical reactions that break down the food and release energy.
- getting food to the cell.
- removing carbon dioxide and other waste products from the cell.

Breathing is the physical means of getting gases in and out of our lungs. The exchange of gases between the atmosphere and our body occurs at the surface linings of the lungs. The lungs are in contact with the circulatory system and blood carries oxygen to the cells.

- 1 The following are all involved in respiration. Rewrite them in the order from outside the body to inside the body.

alveolus artery atmosphere bronchiole bronchus capillary cell trachea

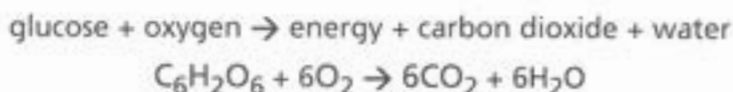
Once oxygen diffuses into the bloodstream, it binds to the protein haemoglobin and oxygen rich blood is carried to the heart. Blood reaches the left atrium, moves to the left ventricle and is then pumped to cells around the body via blood vessels.

- 2 Name the blood vessels that carry blood to the heart.
3 Name the blood vessels that carry blood away from the heart.

Blood vessels that connect various organs to the heart are named to help identify their function. *Femoral* veins and arteries link legs to the heart, whereas *hepatic* veins and arteries link the liver to the heart.

- 4 Identify the organ that is connected to the heart by *renal* blood vessels.
5 Identify the organ that is connected to the heart by *pulmonary* blood vessels.

Many nutrients, for example simple sugars, amino acids, salts and water, enter the bloodstream from the small intestine. Glucose ($C_6H_{12}O_6$) is a simple sugar (a monosaccharide) that is produced when complex carbohydrates (polysaccharides) are broken down in digestion. Glucose dissolved in blood is carried by blood vessels to cells. In a cell, glucose is oxidised in a multi-step process to give energy, carbon dioxide and water.



The process of cellular respiration is more complex than the equation above suggests. *Adenosine triphosphate* (ATP) molecules are also involved. Adenosine triphosphate is used as an energy source for nearly all cellular metabolic processes. Making ATP stores energy, breaking it down releases energy.

Carbon dioxide produced in cellular respiration diffuses into capillaries and then into veins to the heart and from there to the lungs. Most carbon dioxide is transported as the bicarbonate (HCO_3^-) ion. There are enzymes in red blood cells that speed up the formation and breakdown of the bicarbonate ion. This assists in the removal of carbon dioxide from the body.

- 6 What is the major difference between aerobic and anaerobic respiration?

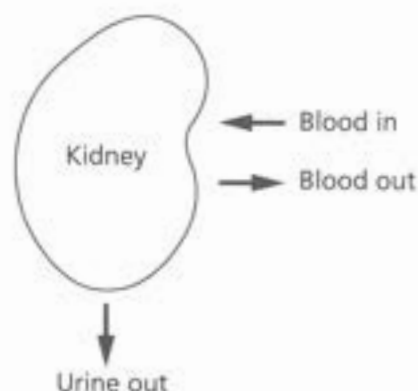
- 7 Identify the organ of the body that has atria and ventricles.
- 8 Identify the organ of the body that carries out gas exchange with the atmosphere.
- 9 Give the full name for ATP.
- 10 In which form is most carbon dioxide carried in the bloodstream?
- 11 Outline the major steps involved in cellular respiration.
- 12 Identify this term: 'An iron-containing pigment that functions in oxygen transport'.

The excretory system

The body needs to get rid of toxic (poisonous) wastes or else we would become ill and die. Excretion removes unwanted substances and helps maintain an optimum concentration of wanted substances in blood and other body fluids.

After food has been digested in the alimentary canal, the waste is passed out of the body via the rectum and anus. The kidneys, lungs and skin are also involved in excretion.

Blood enters the kidneys by a renal artery (a branch of the *aorta*). Once blood is filtered through tiny units called *nephrons*, it leaves by a renal vein, which joins the *vena cava*. In the kidney, useful substances (eg glucose) are reabsorbed and undesirable substances are removed. Blood plasma has a very constant composition, whereas the composition of urine varies considerably. Urine, containing water and waste materials (eg urea and salts), is carried by a *ureter* to the bladder, where it is stored. Urine is discharged from the bladder through the *urethra*.



- 13 How many kidneys does a human have?

Antidiuretic hormone (ADH) decreases the amount of water in urine.

ADH is released by the *pituitary gland* as part of a feed back mechanism to regulate the amount of water in blood and tissues. This process is controlled by the *hypothalamus*. The hypothalamus controls many body functions and helps maintain stable body conditions.

- 14 State the location of the hypothalamus.

The lungs expel carbon dioxide (CO_2) and also water (H_2O). Carbon dioxide turns limewater milky/white and water turns blue cobalt chloride paper to pink. The skin loses water and salts (mainly sodium chloride, NaCl) and urea by perspiration.

State whether the following are true or false:

- 15 When the body needs to conserve water, less ADH is released and urine becomes more concentrated.
- 16 Urea is a nitrogen-based waste substance formed when amino acids are broken down by the liver.
- 17 The vena cava is an artery.
- 18 The kidney removes excess glucose from the blood.
- 19 Hormones in the blood influence the activity of the kidneys.
- 20 When mammals perspire they only lose water through their skin.
- 21 Blood goes into a kidney via a capillary.

Identify the following terms:

- 22 'A canal that carries urine from the bladder.'
- 23 'A single excretory unit of the kidney of a vertebrate.'
- 24 'An organ in humans that lies near the spinal column and that secretes urine.'
- 25 'One of a pair of tubes that carries urine from a kidney to the bladder.'
- 26 'A compound with the formula $\text{CO}(\text{NH}_2)_2$ that is a major component of urine.'

Research

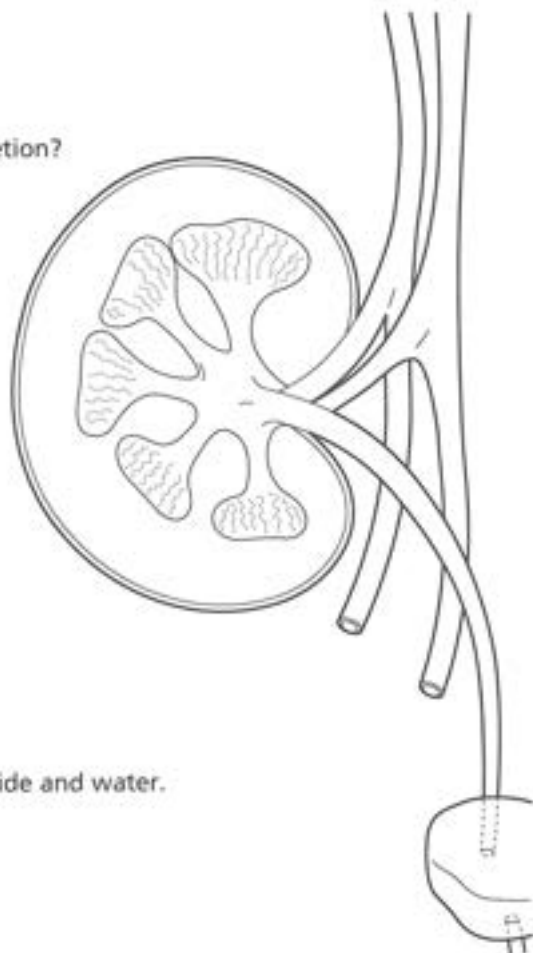
Kidney failure can cause death in humans.

- 27 Briefly describe how dialysis can help people with kidney problems.
- 28 Outline the symptoms of nephritis.

Multiple choice quiz

Choose the best response for each question.

- 29 When a person exhaled through a straw into limewater, the limewater went cloudy.
This shows that:
 - A people breathe out carbon dioxide.
 - B all animals breathe out carbon dioxide.
 - C this person has a respiratory disease.
 - D people breathe out water vapour.
- 30 The excretory system:
 - A regulates the heart beat.
 - B keeps breathing at a steady rate.
 - C removes toxic wastes from the body.
 - D controls our reflexes.
- 31 Which list contains only organs concerned with excretion?
 - A heart, lungs, bladder
 - B kidney, lungs, skin
 - C liver, brain, kidneys
 - D heart, skin, pancreas
- 32 Enzymes are:
 - A toxins.
 - B only found in humans.
 - C very rare.
 - D biological catalysts.
- 33 The word 'renal' refers to:
 - A lungs.
 - B kidneys.
 - C livers.
 - D hearts.
- 34 Glucose ($\text{C}_6\text{H}_{12}\text{O}_6$):
 - A is a monosaccharide.
 - B can be metabolised to give energy, carbon dioxide and water.
 - C a component of polysaccharides.
 - D all of the above.





1 Classification I: Plants and animals

1 Because there are so many; to enable research of specific groups. 2 Orientation of an organism (usually plant) in response to light. 3 Tropism with gravitational orientation (eg plant roots grow down). 4 Eat, reproduce, made of cells, predictable lifespan, usually move or capable of some movement. 5 Carbon dioxide plus water forms glucose plus oxygen. 6 Animal cell: membrane: selects substances that can pass in and out of cell; nucleus: control centre; cytoplasm: where chemical reactions occur; mitochondria: break down substances to provide energy; endoplasmic reticulum: large surface area to store liquids or for chemical reactions; vacuoles: fluid filled sacs. Plant cells also have a cell wall (for rigidity) and green plants have chloroplasts for photosynthesis. 7 autotroph: grass, wattle, tree; heterotroph: cow, sheep, human, fox, rabbit, kangaroo; herbivore: cow, sheep, rabbit, kangaroo; omnivore: human; carnivore: fox (human may be argued to fit in carnivore/herbivore but most eat more than just meat) 8 To be checked. 9 eg have wings, two legs, claws, etc 10 eg fins, live in water, gills 11 eg environment where it is found, behaviour, type of nest/burrow, diet etc 12 ectothermic or poikilothermic 13 Vertebrates have an internal spinal column; invertebrates have no backbone. 14 Their blood can become quite warm in warm conditions. 15 Mammals, birds (endotherms); fish, reptiles and amphibians (ectotherms). 16 Fish cannot live out of water for more than a few minutes (if that) while amphibians can provided water is always nearby. 17 eg swim, hop, slither, glide, fly, crawl 18 Outer skeleton, not within the body eg crab shell. 19 eg both live in water, similar body design; differences: dolphin warm-blooded mammal, suckles young; fish ectotherm, lays eggs. 20 endo – within; ecto – outer; therm – temperature, heat 21 Group of animals of polytypic species, that is slightly different from the norm, usually the result of geographic isolation. 22 Uses less energy to maintain temperature.

2 Classification II: Keys

1 Igon 2 Tapon 3 May be incorrect to presume that different phenotypes are found on each planet, ship may have made several stops along the way or there could be variation on one planet (and no stops were made). 4 eg Keys should enable options to be factored in. 5 To be checked for logical reasoning. 6 eg fresh/salt or brackish water habitat, shell present/absent, fins or flippers. 7 Live both on land and in water, need water to survive, eg newt, frog, salamander. Cannot survive without both so need to be classified separately. 8 Organisms not from the planet Earth.

3 Classification III: Invertebrates

1 Echinoderms, coelenterates, flatworms, segmented worms, molluscs, arthropods. 2 More than one million. 3 Obtains nourishment from a host organism depriving the host of it. 4 eg tapeworm, tick, flea 5 eg starfish, sea urchin 6 eg snail, octopus, oyster 7 liver fluke, tapeworm 8 Outer skeleton, not within body eg crab shell. 9 eight 10 Centipede has one pair of legs per segment, millipede has two. 11 Crustacean – they have gills, antennae and exoskeleton. 12 To be checked. 13 Have some characteristics in common with insects, eg body shape, antennae, 'nippers'. 14 centipede: carnivore; millipede: herbivore

4 Classification IV: Vertebrates

1 Aquatic: to do with water; criteria: standards; cartilage: gristle; fins: appendage of fish; gills: breathing organs of fish; mucus: slippery secretion; bladder: membranous sac; metabolic: body's chemical processes. 2 Endotherm – animal that has a steady internal temperature due to control of metabolism. Ectotherm – animal that has a body temperature that varies with external conditions. 3 snake, lizard, turtle, crocodile etc 4 animals 5/6 frogs/toads 7 moist 8 heart 9 eggs 10 shells 11 water 12 gills 13 develop 14 plants 15 air 16 eating 17 not 18 backbone 19 day 20 salamander, newt, caecilians etc 21 No, temperature is too cold. 22 Hibernates: pass the winter in a torpid/inactive/resting state. Estivate: pass the summer in a torpid/inactive/resting state. 23 eg Survive extreme temperatures and food shortages; conserve energy. 24 They have always been well adapted to their environment.

5 Mammals – monotremes and marsupials

1 Australia, New Zealand and islands of southwest Pacific. 2 Warm-blooded (endotherm), suckles young, usually has fur. 3 Environment in which an organism lives, all that makes up its home. 4 Combined tissue enables body to recognise 'self'; shared hormones help with this unity. 5 A particular hormonal level. 6 Born before the completion of the usual term of pregnancy. 7 Active at night. 8 Time the young is developing within the mother. 9 Eutherians: complex and longer term connection between young and placenta; young are born in more developed state than marsupials. Marsupials: young are born underdeveloped and continue development within a pouch. Monotremes: lay eggs which contain sufficient nutrient for young to grow to a certain stage after which mother suckles it on milk. Note all groups suckle young. 10 quoll, dunnart 11 echidna, numbat 12 scavenger 13 Eutherian: human, rodent, apes and monkeys, cat, dog; Marsupial: kangaroo, wallaby, Tasmanian devil, wombat, bilby, glider, possum; Monotreme: platypus, echidna 14 a) eg both produce undeveloped young, found on similar continents b) both lay eggs 15 To be checked for logical argument. 16 Marsupial mice (dunnart, antichinus) share physical resemblance and ecological lifestyle to house mouse. 17 Quoll: native cat; Antichinus and dunnart: marsupial mice 18 To be checked. There is no universally agreed sequence and opinions vary greatly. 19 kangaroo

6 Species that have become extinct

1 Fossil: remains, impression of a once living organism; eliminated: removed; thriving: doing well, healthy, large populations; extinctions: no longer have living specimens; overhunting: hunting at unsustainable levels 2 emu 3 Aborigine 4 Driven over the cliff in herds; inefficient, wasteful. 5 spear, boomerang 6 Large-scale extinction occurred twice. 7 dwarfing 8 marsupial 9 smoulder 10 mammoth 11 drought 12 climate 13 Sarcophilus harrissi 14 Thylacinus cynocephalus

7 Revision assignments

1 To be checked. 2 Males venture across wider ranges than females, more males than females or females are quicker/more alert than males, males may prefer habitat near roadside. 3 To be checked for logic, eg females more

timid/less adventurous. 4 The other species are attracted to the carcass and get hit by cars when scavenging. 5 Flightless, large, body shape, etc. 6 a Against: eat grass that could be used by sheep/cattle; ruin fences. b For: kangaroos cause little damage to habitat. c Against: they can be a hazard on the road and cause damage to cars if hit. d For: natural food source. 7 eg manage lands, reduce fire hazard, cause fire-resistant species to thrive, encourage seed germination of certain plants. 8 Destruction of areas of native flora and fauna, possibility of causing further damage. 9 The NSW Wildlife Information and Rescue Service (WIRES) is the largest wildlife rescue organisation in Australia. (See website <http://www.wires.org.au/about/about.htm>)

8 The discovery of the cell

1 Experiments with the relationship between matter and energy 2 Studies functions and interactions between organs and tissues 3 Animal Science 4 Plant Science 5 Studies living organisms, plant and animal, and the processes that maintain life for each. 6 To be checked. 7 *Schwann cell*: specialised (glial) cells enveloping central nervous system axons. *Purkinje cell*: large neurones in cerebral cortex. *Purkinje fibre*: modified cardiac muscle fibres. 8 Cell is the smallest form of life, cells can only arise from pre-existing cells, cell is basic primary structure of living things, not all components of living organisms contain cells. 9 eg a single cell (in the right conditions) can survive in its own right, taking in nutrients etc. Organelles cannot survive independently and atoms and molecules are inorganic. Student to show that subcellular level cannot exist alone so is not an independent form of life. 10 *Plant*: bark; *Animal*: hair, nail (do not contain living cells). 11 eg *unicellular*: protozoa; *multicellular*: metazoa 12 Requires many cells to increase size, one cell could not cope or become large enough. 13 Organic matter is made of elements. Elements and molecules are not living matter. 14 atom, cell, tissue, organ, organ system, multicellular organism 15 organ: functional unit of anatomy, made of tissue; organelle: functional part of a cell. 16 chlorophyll: green pigment found in plants; chloroplasts: plastid containing chlorophyll. 17 molecule: smallest particle of a compound that contains compound's properties; compound: two or more elements mixed to form a substance. 18 true 19 true 20 false 21 false 22 true 23 true 24 true 25 true 26 false 27 To be checked.

9 Microscopes I

1 No artificial magnification used. 2 Teacher to check using dates in text. 3 A simple microscope has only one lens; a compound microscope had two or more lenses, allowing greater magnification. 4 anthrax 5 1 to 10 micrometres (10^{-6} m) 6 Cell structure, chromosomes and mitosis. 7 anatomy 8 mycology 9 pathology 10 histology 11 virology 12 phycology 13 embryology 14 ornithology 15 cytology

16 Ocular lens	Place eye to look through this
Arm	Pick up microscope by this part
Objective lens	Lens closest to specimen
Revolving nosepiece	Has lenses of different magnification

Mirror	Reflects light onto specimen
Condenser lens	Focuses light from mirror onto specimen
Fine focus	Brings specimen into sharp focus
Base	Lowest part, support for microscope
Course focus knob	Used to get specimen approximately in focus
Stage	Microscope slide is placed on this

10 Microscopes II

magnified, light, electron, microscope, compound, lenses, eyepiece, objective, specimens, slides, focus, stained, thousands, resolution, small, millionths, smaller, viruses, molecules, microorganisms, century, Hooke, cells, botanist, plants, division, dyes

11 Structure of plant and animal cells

1 autotroph: makes own food; heterotroph: relies on other organisms for food 2 cellulose 3 eg vacuoles, mitochondria 4 nucleus 5 Adenosine triphosphate 6 chromosome 7 Bacteria, spirochetes, cyanobacteria (blue-green algae), actinomycetes etc. 8 Part of a cell; 'little organ'. 9 Protects and supports cells, makes plants rigid. 10 They have other means of support eg exoskeleton, endoskeleton. 11 Only allows passage of some substances eg during osmosis. 12 Starch grain, plastid, chloroplast, cell wall. 13 Changing colour of an object to make it more visible under a microscope. 14 one cell 15 killing 16 many cells 17 Cytoplasm, mitochondria, nucleus and plastids of a cell. 18 Mobile ones may be difficult to view. 19 Much greater magnification.

12 Cell components

1 DNA	material that forms genes
deoxyribose	sugar found in DNA
gene	smallest unit of heredity
chromosomes	structures containing DNA
nucleotide	unit containing a sugar, a base and a phosphate
RNA	nucleic acid similar to DNA
nucleolus	'little nucleus'

2 Controls transport of materials. 3 Attracted to water. 4 Repelled by water. 5 fat/oil 6 Very small hole for absorption, perspiration etc. 7 Cell is complete entity; organelle is a specific part of a cell; cytoplasm is the material outside the nucleus excepting the organelles. 8 intra - inside; inter - between 9 Interconnects all the parts of a cell. 10 Check for logical argument. 11 Cell itself could die. 12 Biological catalyst. 13 glucose 14 oxygen 15 carbon dioxide 16 water 17 Regulates traffic into and out of cell. 18 Gives instructions, holds genetic material. 19 Converts food to energy. 20 Site of chemical reactions. 21 Modifies proteins. 22 Breaks down materials. 23 Enzyme found in saliva and tears.

13 Substances in cells I

1 End in '-ose'. 2 Yes, carbon is present and H:O ratio is 2:1. 3 Mono - one, di - two, poly - many. Refers to the number of rings of atoms present in sugar. 4 Carbon, hydrogen and oxygen. 5 Check diagram, $C_6H_{12}O_6$ 6 Glucose is more quickly metabolised than other sugars. 7 Benedict's solution

changes from blue to orange/red. 8 Iodine solution changes from yellow/brown to blue/black. 9 It is stored in cell walls and used for support. 10 Carbon, hydrogen and oxygen. 11 Carbon, hydrogen and oxygen. 12 Obesity, heart disease. 13 Fat is a solid at room temperature, oil is a liquid. 14 water-proofing. 15 Chemical produced by a gland that is carried in blood to act elsewhere in the body. 16 Turns paper greasy/transparent. 17 nitrogen. 18 no. 19 growth, repair, energy. 20 If protein is present, Biuret solution turns from blue to purple. 21 false.

14 Substances in cells II

1 Mineral	Symbol	Use
Calcium	Ca ²⁺	Needed for healthy teeth and bones.
Iodine	I ⁻	Used by thyroid gland to synthesise hormones.
Iron	Fe ³⁺	Essential part of haemoglobin, used to transport oxygen through the body.
Magnesium	Mg ²⁺	Helps nerve cells to work properly.
Phosphorus	PO ₄ ³⁻	Important in metabolism of cells.
Potassium & Sodium	K ⁺ Na ⁺	Transfer electrical impulses in and between nerve cells.

2 Dairy products (milk, cheese, yoghurt). 3 Not enough sodium in the blood. 4 Too much sodium in the blood. 5 Not enough potassium in the blood. 6 Too much potassium in the blood. 7 k 8 d 9 j 10 i 11 i 12 h 13 c 14 a 15 b 16 e 17 f 18 Add silver nitrate solution to get a white precipitate. 19 The greater part of a solution which can dissolve the other parts, eg haemoglobin, carbonates, proteins. 20 solute. 21 Solute particles move through semipermeable membrane to mix more evenly with water molecules, equalising the concentration of the substance. 22 Solvent particles (ie water molecules) move through semipermeable membrane to areas of higher solute concentration, therefore equalising the concentration of the substance. 23 Caused by motion. 24 Not moving in a predictable or uniform manner. 25 The science of plants. 26 The fertilising element of flowering plants (yellow powdery grains). 27 So oxygen can be transported around the body, enabling various chemical reactions to occur. 28 Travelled with Matthew Flinders on board the *Investigator*.

15 Bacteria

1 bacterium. 2 streptococci. 3 cocci. 4 staphylococci. 5 diplococci. 6 Cocci: plaque, tooth decay; diplococci: gonorrhoea, meningitis, pneumonia; staphylococci: toxic shock syndrome, pneumonia, septicaemia; streptococci: rheumatic fever, scarlet fever, tonsillitis etc. 7 movement. 8 one million (10⁶). 9 They can't move under own power. 10 shape. 11 Because they are so tiny, they leave very few fossils. 12 Spiral, curve, 'comma'. 13 Bacteria with slender spirally undulating shape, includes those causing syphilis and relapsing fever. 14 Allowing only some substances to pass through. 15 Sexual reproduction requires male and female organism; leads to genetic variation. 16 B, E, C, D, A. 17 2⁴⁸, 28.1 x 10¹⁴. Not enough food, some bacteria will die etc. 18 Teacher to check, should include cell wall, nuclear material, cytoplasm, cell membrane. 19 environment:

surroundings both living and non-living; *hydrothermal vents*: hot water springs; *anaerobic*: lives in absence of air; *metabolise*: use/break down; *differentiate*: make difference between; *symbiont*: lives on living organisms; *saprophyte*: lives on dead organisms; *unpalatable*: not fit to eat; *pathogenic*: causes disease; *antibiotic*: kills bacteria.

16 Plants: multiple choice test

1 C 2 D 3 C 4 B 5 B 6 A 7 D 8 B 9 A 10 C 11 B 12 A 13 A 14 C 15 D

17 Conditions needed for life

1 Move, reproduce, respire, take in nutrients, excrete, respond to stimuli etc. 2 *Photosynthesis*: water plus carbon dioxide plus energy produces glucose and oxygen. *Respiration*: glucose combining with oxygen produces energy plus water plus carbon dioxide. 3 Photosynthesis only occurs in sunlight, but respiration occurs all of the time. 4 Nitrogen, phosphorus, potassium etc. 5 Autotroph – makes own food; heterotroph – needs other organisms for food. 6 Teacher to check, but statement has merits. 7 dehydration. 8 vitamins. 9 Contains no energy. 10 Urine, sweating, exhaling. 11 Needed to keep body systems working. 12 true. 13 true. 14 false. 15 false. 16 true. 17 832.5 kJ. 18 We gain weight. 19 eg bread, potatoes, pasta, rice. 20 Glucose is metabolised very quickly, giving energy. 21 proteins. 22 Growth, repair and energy. 23 Starch – iodine, glucose – Benedict's solution. 24 A sugar. 25 Fat is solid at room temperature, oil is liquid. 26 Energy store, insulation. 27 Sucrose takes longer to digest. 28 Plants will use up oxygen during respiration. 29 Body systems continue, eg breathing, heart beating, brain function etc. 30 They are growing at a fast rate, generally more active, brain absorbing new information etc. 31 Muscles keep toned, heart healthier, good blood circulation, mental attitude etc. 32 You need a balanced, varied diet to be healthy. It can still be low in fat. 33 For: healthier to avoid animal fats, less pressure on environment. Against: not a balanced diet, vegetarian diet may not supply all substances needed by the body. 34 Vegetarians eat some animal products eg milk, cheese, vegans don't eat any. 35 body fat. 36 extremely overweight. 37 One who studies food and diet. 38 eggs, fatty meat, avocado etc. 39 heart disease, obesity. 40 canning, pickling, fermenting, freezing, drying.

18 Ectotherms and endotherms

1 Temperature that suits the organism best. 2 Temperature of surroundings. 3 Aquatic – to do with water; terrestrial – to do with land. 4 Temperatures on land vary more than those in water. 5 Reptiles, eg blue-tongued lizard, frill-necked lizard, snakes etc. 6 Various snakes. 7 Various moths. 8 Teacher to check, should show exponential rise. 9 As temperature rises, metabolic rate increases exponentially. 10 Animals might not exist in these temperatures. 11 Size and colouring reduces heat loss. 12 Fur, fat as insulation. 13 Can hunt or escape from enemies more readily. 14 Most active at night. 15 Lose heat through evaporation. 16 Radiates excess heat. 17 *Ectotherms*: barramundi, blue-tongued lizard, Bogong moth, frill-necked lizard, goanna, green tree frog, shark, white tip spider. *Endotherms*: bats, echidna, emu, platypus, sugar glider. 18 Difficult to lose heat in warmer-

than-body surroundings. 19 To warm up hive, to circulate cooler air. 20 hibernation 21 metabolism 22 false 23 true 24 false 25 false 26 true 27 false 28 false 29 true 30 true 31 true 32 To spend the winter hiding in a dormant state. Metabolism slows, body temperature drops, energy use is minimised. 33 seals, sealions, polar bears, penguins etc 34 eg air conditioning, shaded housing, fans, clothing etc 35 wasps, hornets

19 Homeostasis

1 *Endotherm*: body temperature varies with external temperature (cold-blooded/ poikilothermic). *Ectotherm*: body temperature remains constant (warm-blooded/ homeothermic). 2 Fit together to create a different molecule. 3 Site where enzyme binds to a molecule. 4 A = energy, B = activation energy (catalysed), C = activation energy (uncatalysed), D = Energy released by reaction 5 enzyme 6 true 7 true 8 true 9 false 10 false 11 true 12 As temperature increases so does reaction rate. Once a maximum temperature is reached, reactions stop. 13 37.4°C (about 98°F) 14 Should have 'bell curve' similar to figure 3. 15 Increases with increasing substrate concentration to a maximum and then levels off. 16 substrate 17 pH 18 denature 19 true 20 true 21 false 22 false

20 Human systems I

1 atoms, molecule, cell, tissue, organ, organ system, organism 2 Artery carries more than vein which carries more than capillary. Largest diameter carries more blood.

3 See table 1 (below). 4 Regulate blood flow, prevent back flow. 5 Forward flow of blood pushes them open, back flow pushes them shut. 6 heart 7 Transport nutrients, remove waste. 8 four 9 There is more pressure on them. 10 Lymph, tissue fluid, bathes cells, links arteries and veins etc. 11 William Harvey (1578–1657) explained workings of heart valves, arterial pulse, pulmonary circulation, venous valves etc. 12 carbon dioxide, urea etc 13 See table 2 (below). 14 antibodies 15 plasma 16 red blood cells (erythrocytes) 17 enzymes 18 hormones 19 nucleus 20 opaque 21 37.4°C (about 98°F) 22 7.4 23 Brings blood to skin's surface to cool it. 24 artery, vein, capillary 25 Spins at high speeds to separate the heavier red blood cells from plasma, which sits on top as a liquid. 26 D 27 C 28 A 29 A 30 B

21 Human systems II

1 atmosphere, trachea, artery, bronchus, bronchiole, alveolus, capillary, cell 2 veins 3 arteries 4 kidney 5 lungs 6 Aerobic is in the presence of oxygen, anaerobic is in the absence of oxygen. 7 heart 8 lungs 9 adenosine triphosphate 10 gas 11 To be checked, include glucose + oxygen gives energy plus carbon dioxide plus water. Also mention role of ATP. 12 haemoglobin 13 two (normally but can cope with one) 14 brain 15 true 16 true 17 false 18 false 19 true 20 false 21 false 22 urethra 23 nephron 24 kidney 25 ureter 26 urea 27 Filters and purifies waste from blood. 28 Inflammation of kidneys, acute and chronic forms. 29 A 30 C 31 B 32 D 33 B 34 D

TABLE 1:

Blood vessel	Properties	Role
Arteries	Wide, thick-walled, bright red	Carry blood containing dissolved oxygen, food and waste products.
Veins	Narrow, thin-walled, contain valves to stop backward flow of blood	Carry blood to the heart.
Capillaries	Smallest, thinnest vessel, branch off arterioles and rejoin to form veins	Pick up dissolved food from digestive system, deliver carbon dioxide and take away oxygen from lungs; allow food, water & oxygen to pass into cells, and waste products to pass out of cells.

TABLE 2:

Component of blood	Properties	Uses
Plasma	Pale, yellowish liquid, 90% water	Contains antibodies, enzymes, hormones, dissolved food to supply to cells, and waste materials removed from cells.
Red blood cells	Concave, disc-shaped cells with no nucleus; made in bone marrow	Contain haemoglobin and carry oxygen around the body.
White blood cells	Opaque, larger than red cells, irregularly-shaped nuclei	Important in body's defences against illness; make antibodies and can attack bacteria.
Platelets	Minute, flattened bodies of protoplasm	Important in clotting of blood.

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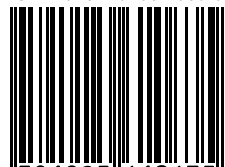
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